Metabolomics Course

- Metabolomics --> why?
- Data acquisition
- How is the data?
- Explain data parts (terms), the ones useful and add the vizualization in the hands-on
- Pre-processing steps: peak centroiding, peak detection, RT alignment, feature matching (steps in OpenMS and in xcms)

- **Add summaries from time to time**

Introduction to metabolomics

Metabolome

Complete set of small molecules (usually <1500 Da) present in a biological system at a given time

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Sources: exogenous, endogenous, microbiome-derived

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- **Sources:** exogenous, endogenous, microbiome-derived
- **Classes**
  - Carbohydrates
  - Nucleotides
  - Amino acids
  - Lipids
  - …

Introduction to metabolomics

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- Classes
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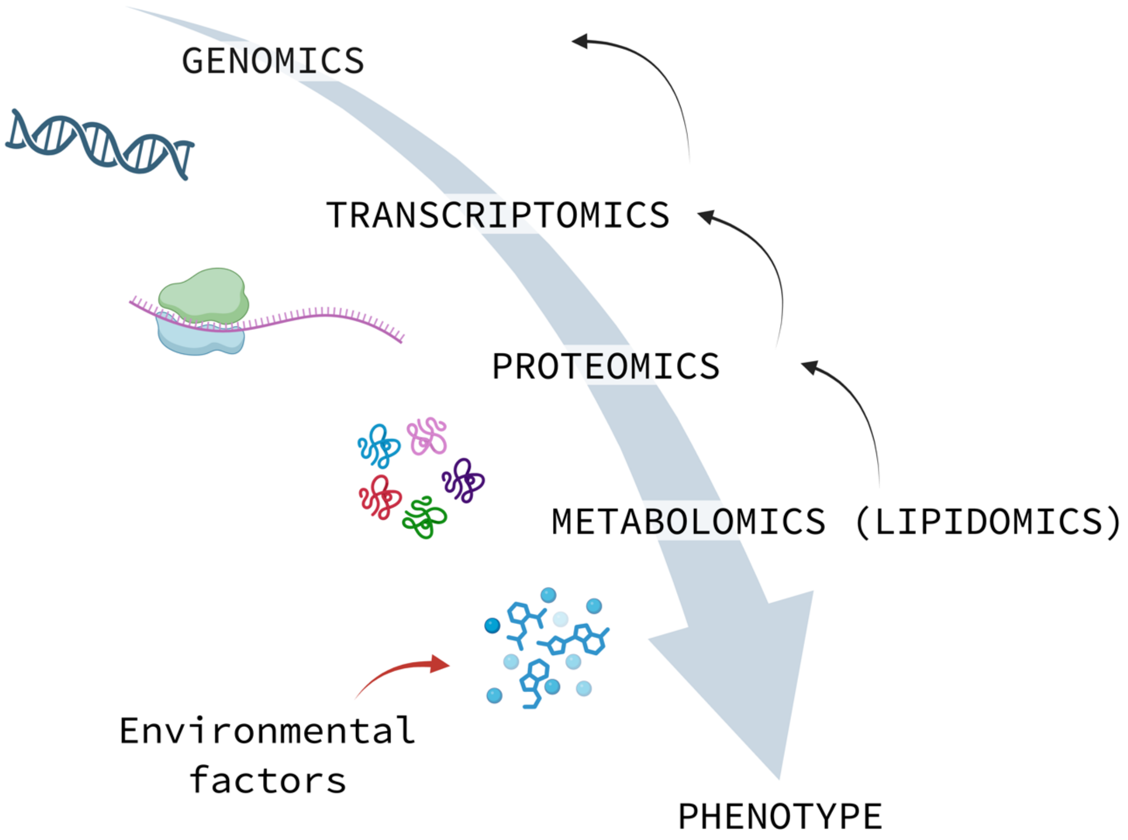
  - …

Lipidome (Lipids)

- The total lipid content in a biological system
- Hydrophobic or amphipathic small molecules

Introduction to metabolomics



Introduction to metabolomics

Applications

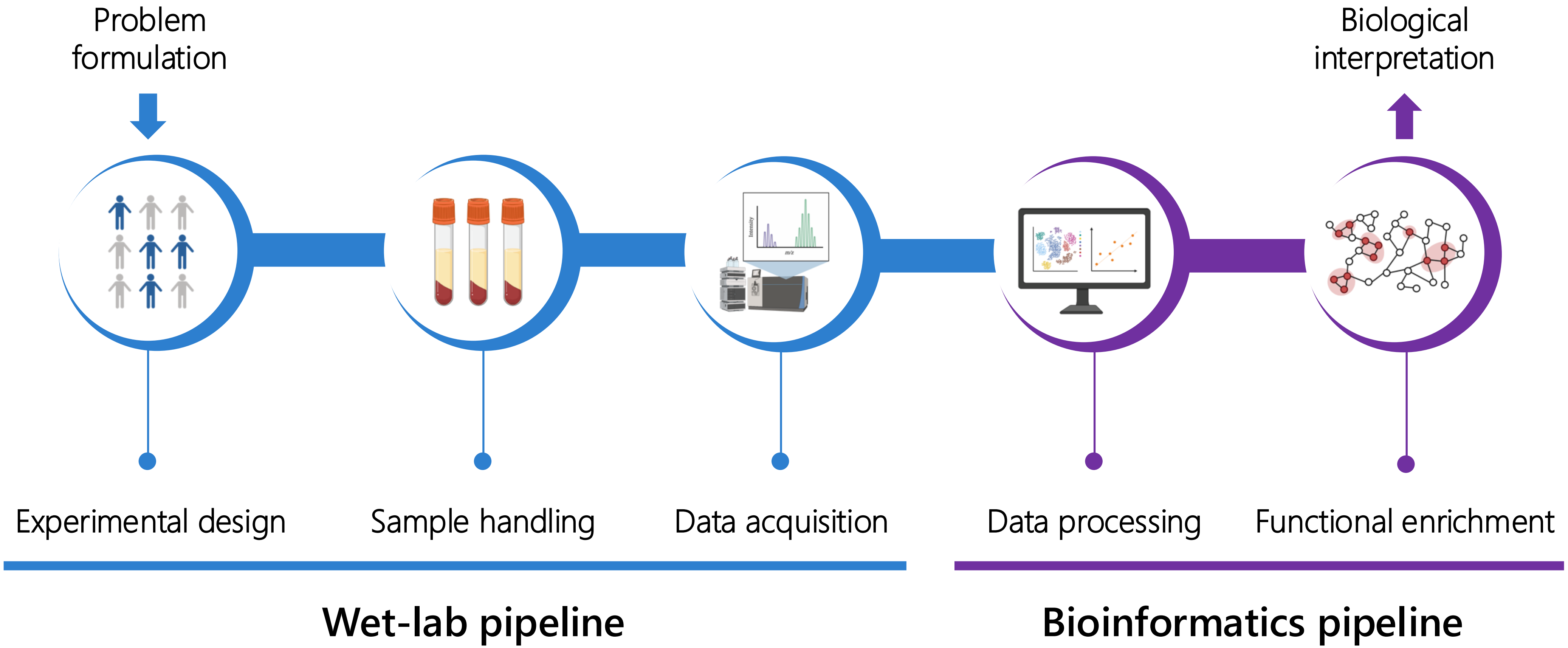
1. Discovery of new disease biomarkers
2. The understanding of molecular mechanisms in pathophysiological processes
3. Nutrition research
4. Drug discovery
5. Fermentation
6. Environmental Monitoring and Ecotoxicology

Introduction to metabolomics

Study types

1. Untargeted
2. Targeted
3. Semi-targeted

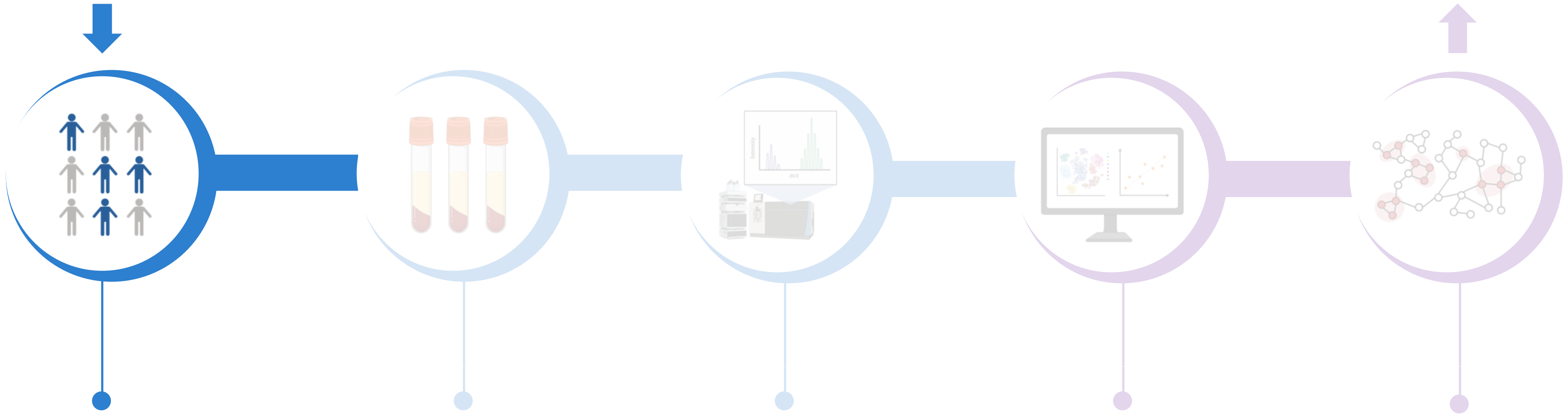
Metabolomics pipeline



Metabolomics pipeline

Problem formulation

Biological interpretation



Experimental design

Confounding factors:

- Analytical
- Physiological

Sample handling

- Collection, storage and preparation
- Tissues, cells, biofluids
- Isolate molecules of interest.

Data acquisition

- LC-MS
- GC-MS
- NMR

Data processing

- Pre-processing
- Data filtering and normalization
- Statistical analysis
- Peak annotation

Functional enrichment

- Over-representation analysis
- Functional class scoring
- Pathway topology

Metabolomics pipeline

Experimental design

Deal with confounding factors that could correlate or mask the biology under-study:

- Analytical: batch effects, injection order, etc.
- Biological: sex, age, body mass index, etc.

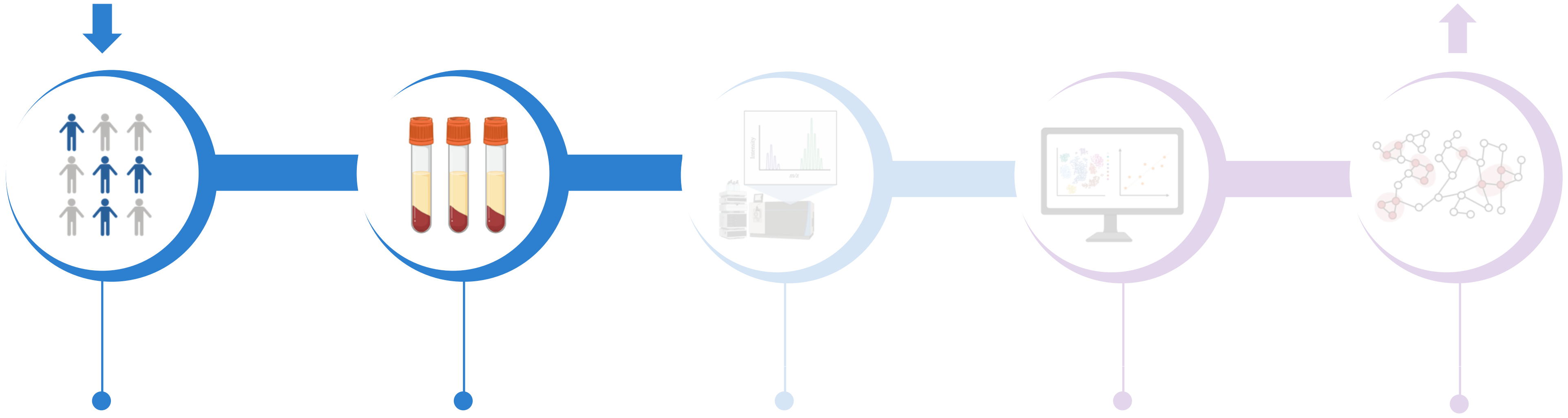
Biological samples randomization based on their physiological characteristics

Quality control (QC) samples are injected at the start, during and end of the acquisition to stabilize the signal and correct for technical variability.

Metabolomics pipeline

Problem formulation

Biological interpretation



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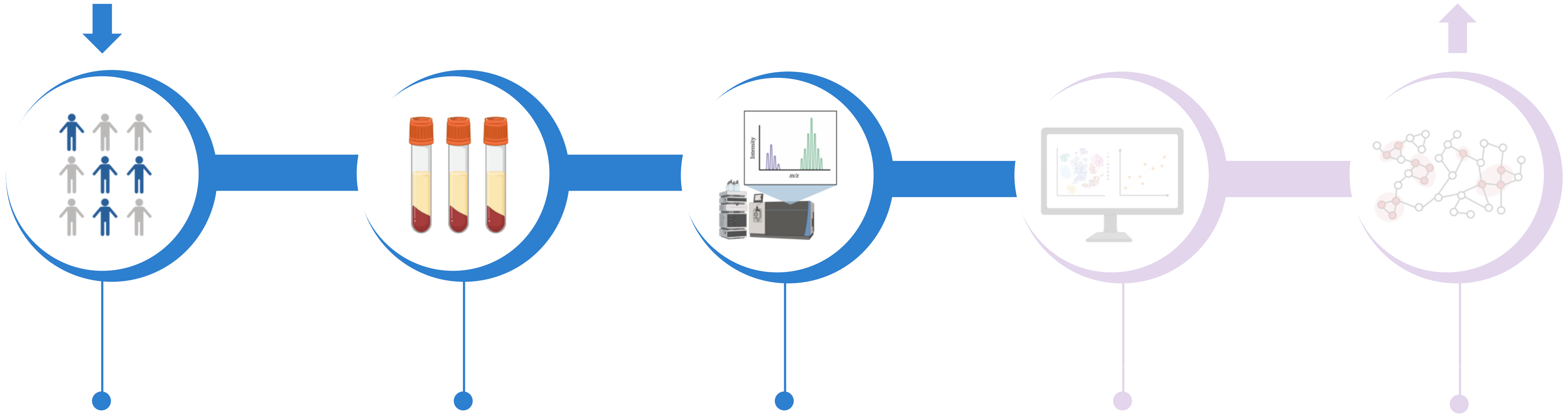

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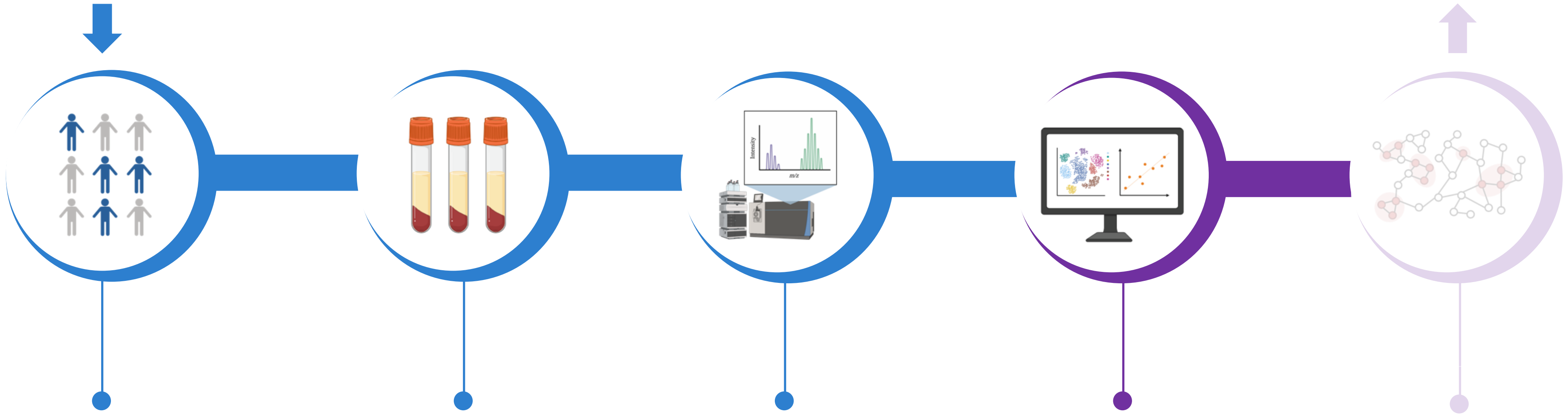

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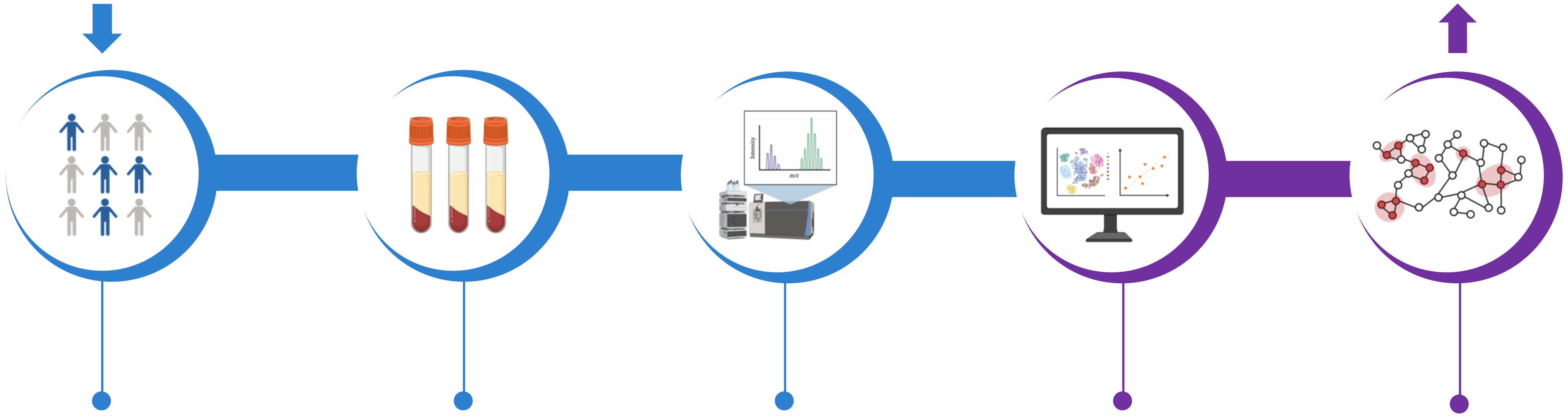

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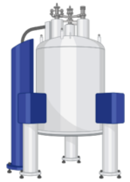
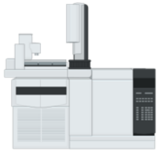
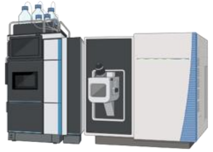
Metabolomics data acquisition



Nuclear Magnetic Resonance (NMR)



Gas chromatography-Mass spectrometry (GC-MS)



Liquid chromatography-Mass spectrometry (LC-MS)

Metabolomics data acquisition

Nuclear Magnetic Resonance (NMR)

STRENGTHS

1. Non-destructive
2. Easily quantifiable
3. No prior separation required
4. Identification of novel compounds
5. High reproducibility

LIMITATIONS

1. Low-sensitivity
2. High detection limit
3. Large sample volumes

Metabolomics data acquisition

Gas chromatography-Mass spectrometry (GC-MS)

STRENGTHS

1. More mature and cost-effective
2. Excellent separation reproducibility
3. Availability of spectral libraries facilitates metabolite identification

LIMITATIONS

1. Sample derivatization required
2. Smaller coverage
3. Longer experimental times

Metabolomics data acquisition

Liquid chromatography-Mass spectrometry (LC-MS)

STRENGTHS

1. Broader range of metabolites
2. Simpler sample preparation
3. High sensitivity

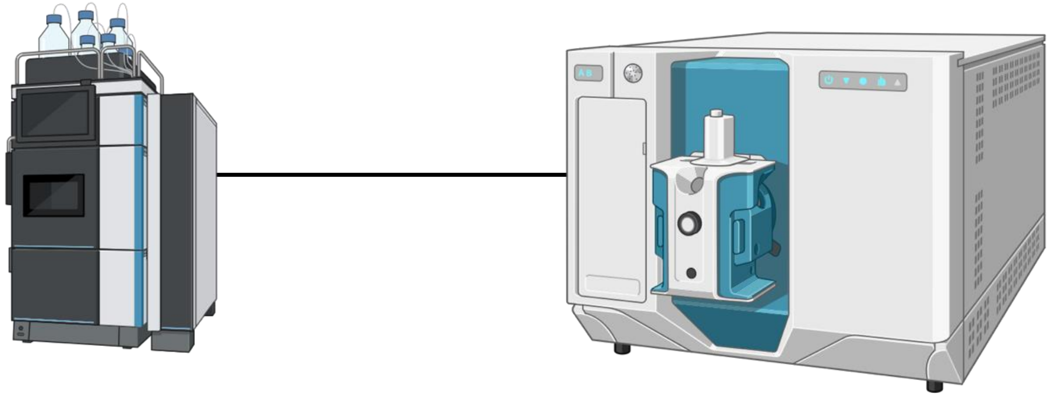
LIMITATIONS

1. Higher cost
2. Lower reproducibility
3. Reduced compatibility with volatile compounds

Preferred choice in metabolomics: **more than 70%** of the metabolomics studies were using LC-MS in 2022 .

Metabolomics data acquisition

Liquid chromatography-Mass spectrometry (LC-MS)



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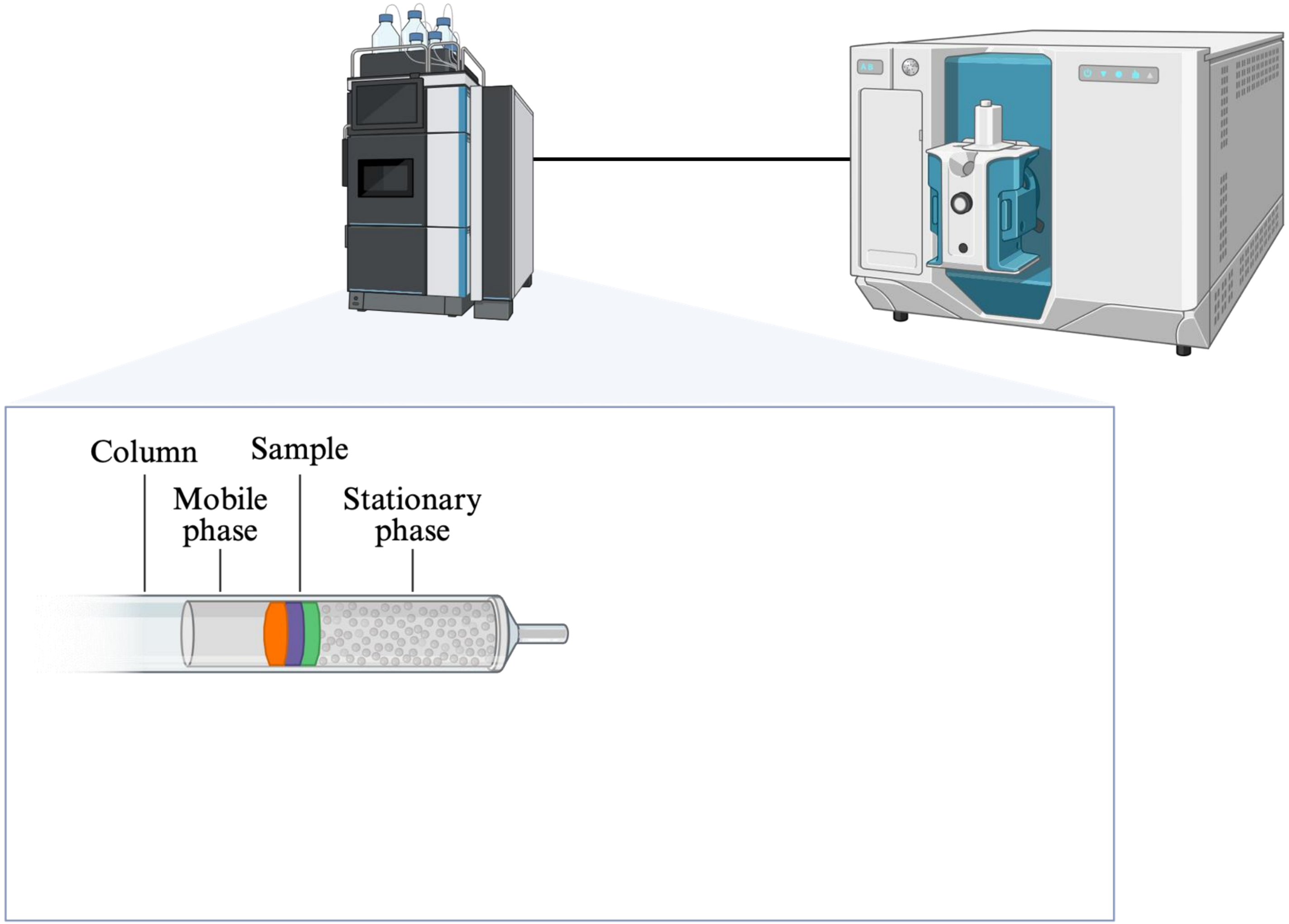

Metabolomics data acquisition

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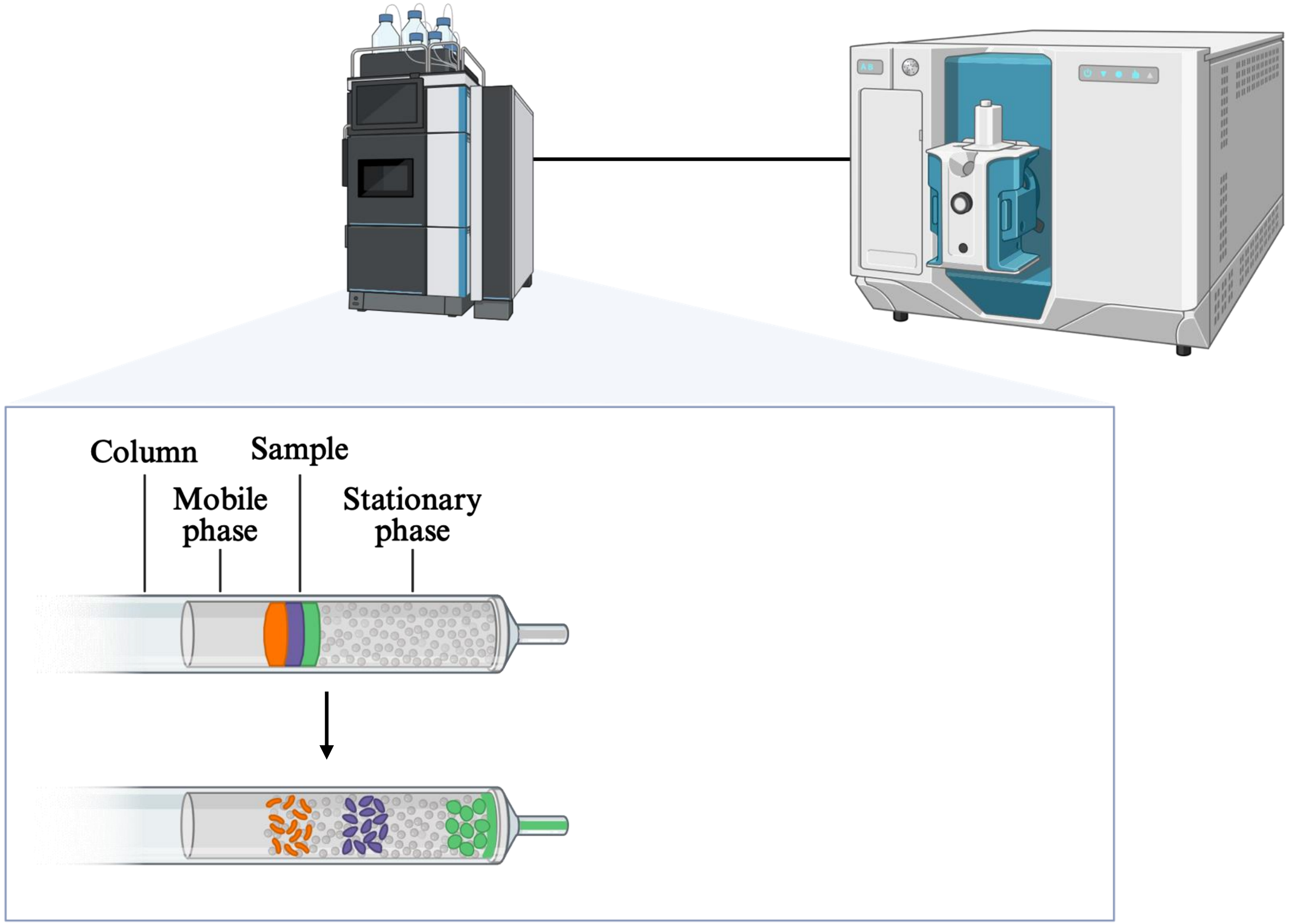

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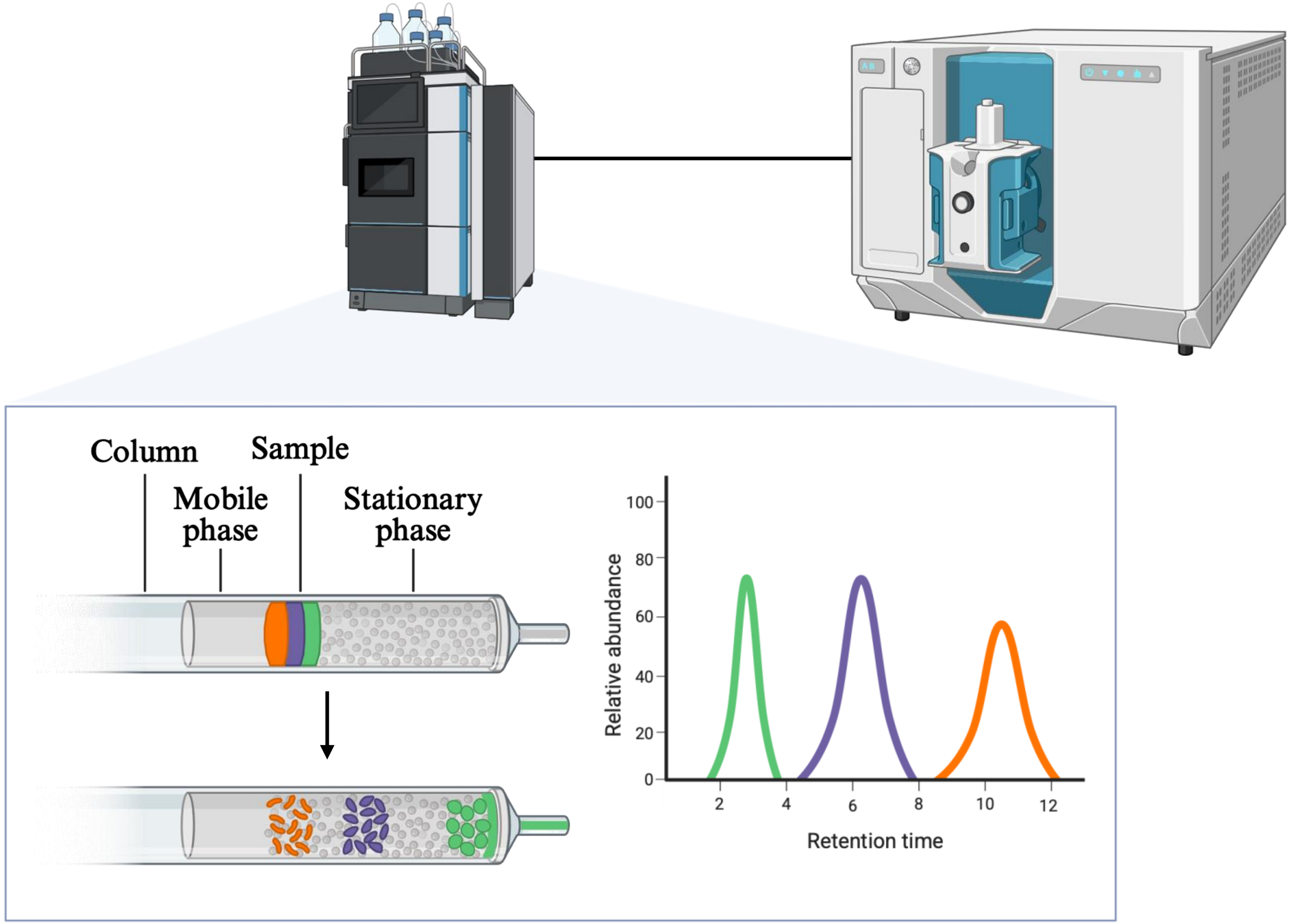

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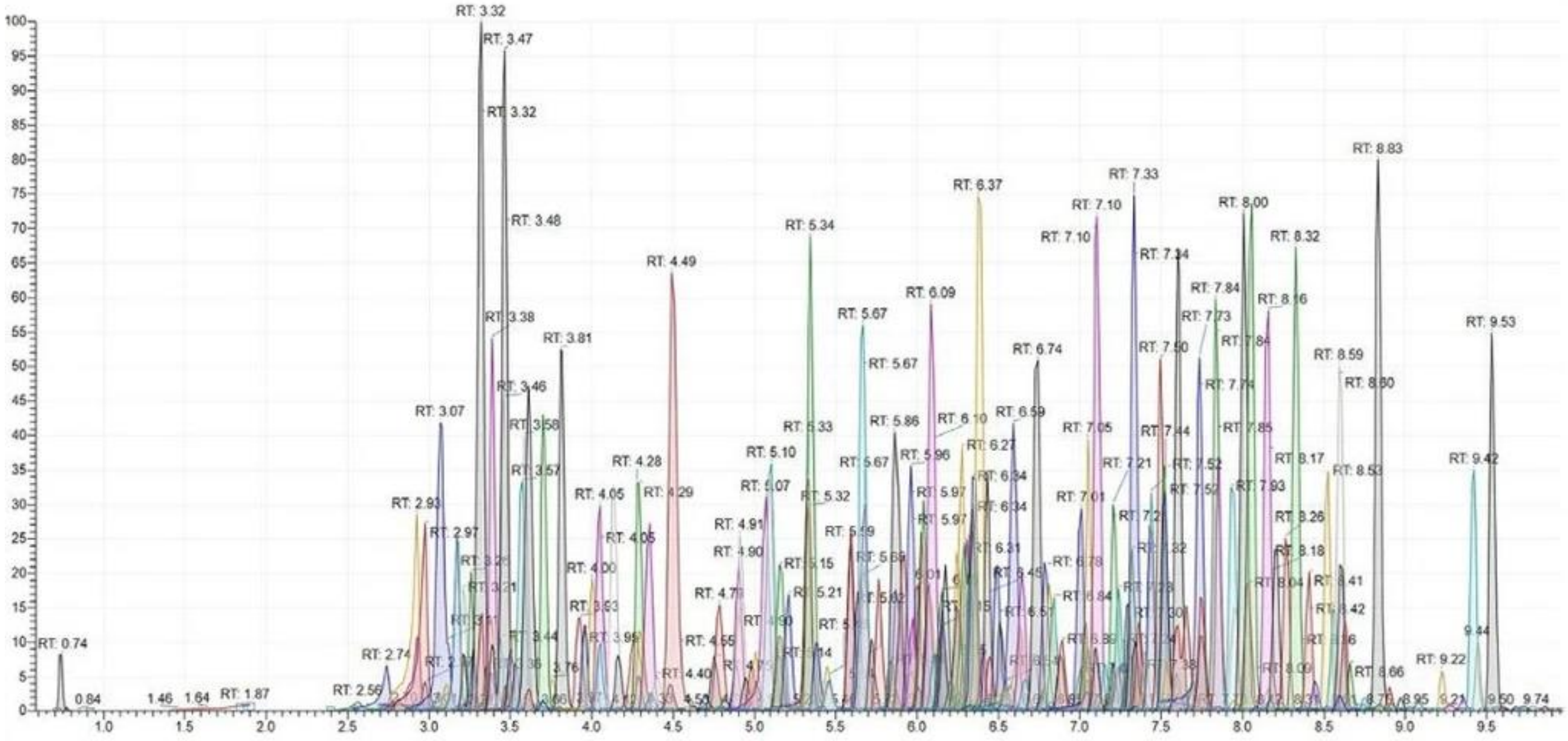

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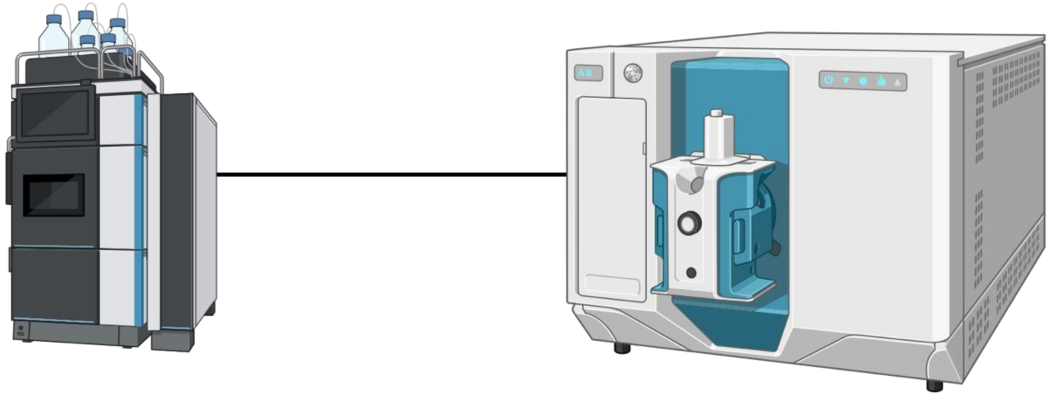

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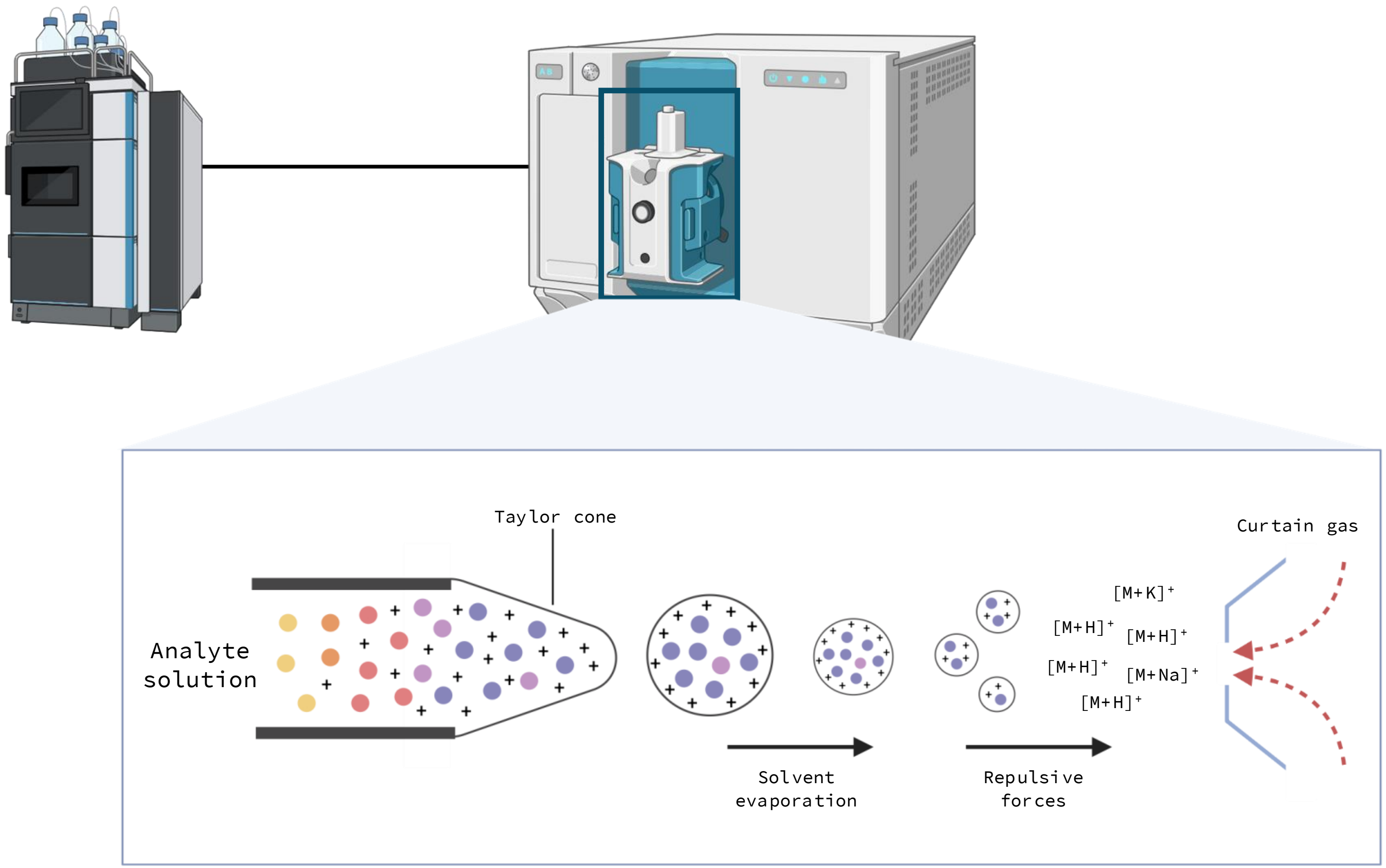

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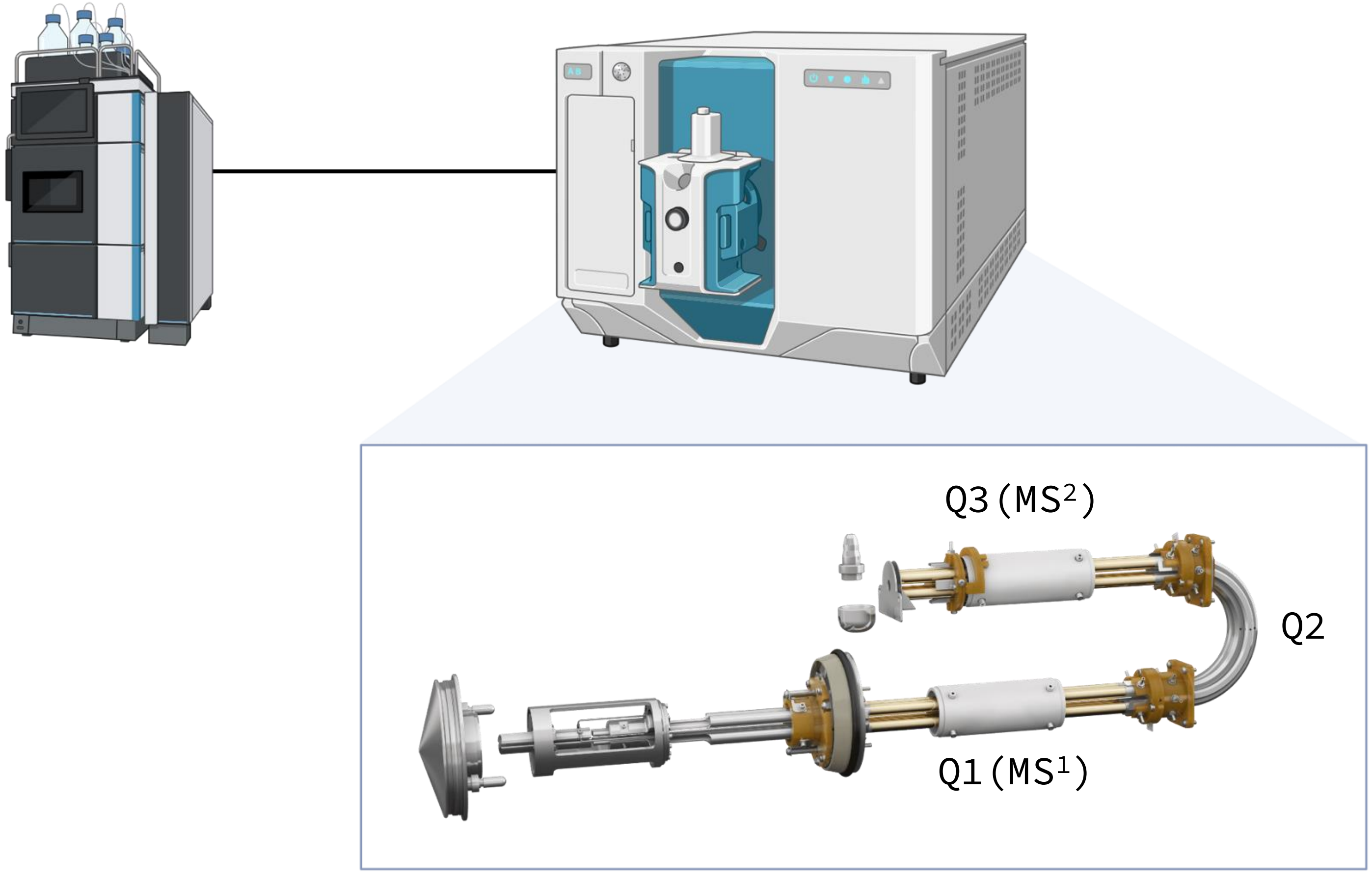

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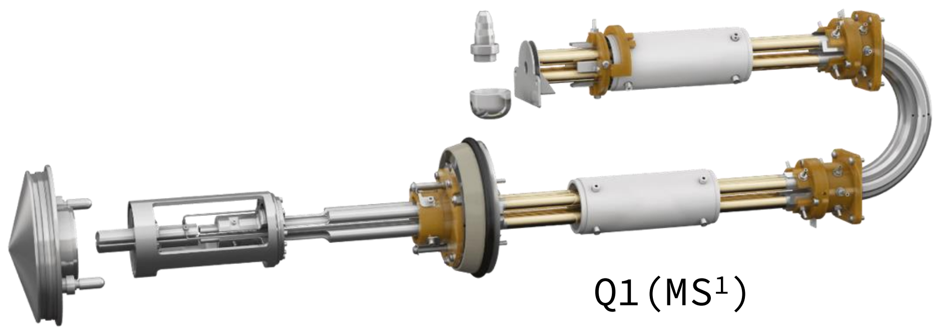

Metabolomics data acquisition

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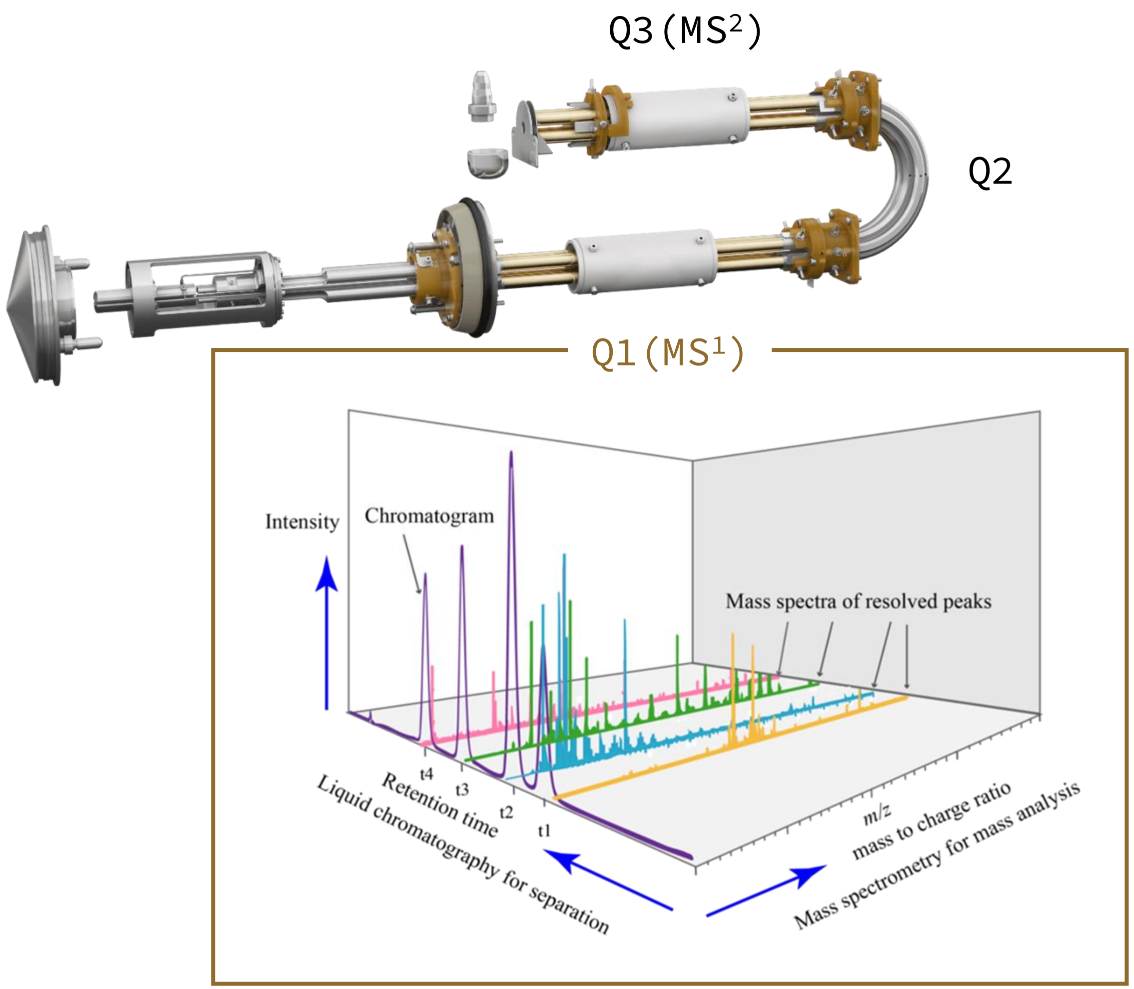
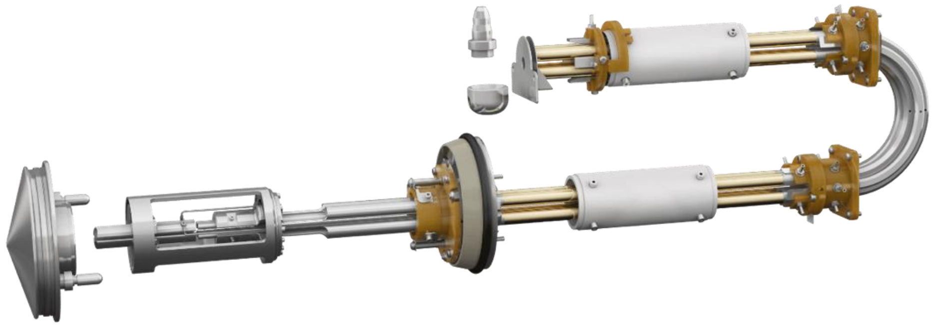


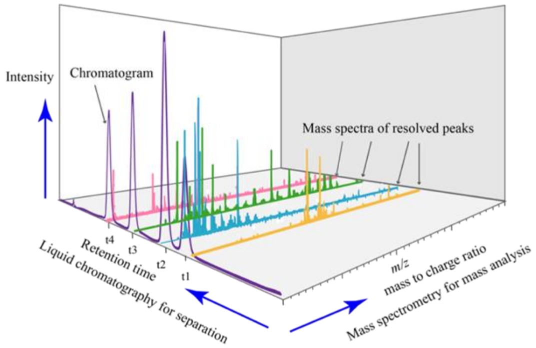
Metabolomics data acquisition

Liquid chromatography-Mass spectrometry (LC-MS)

Q3 (MS^2)

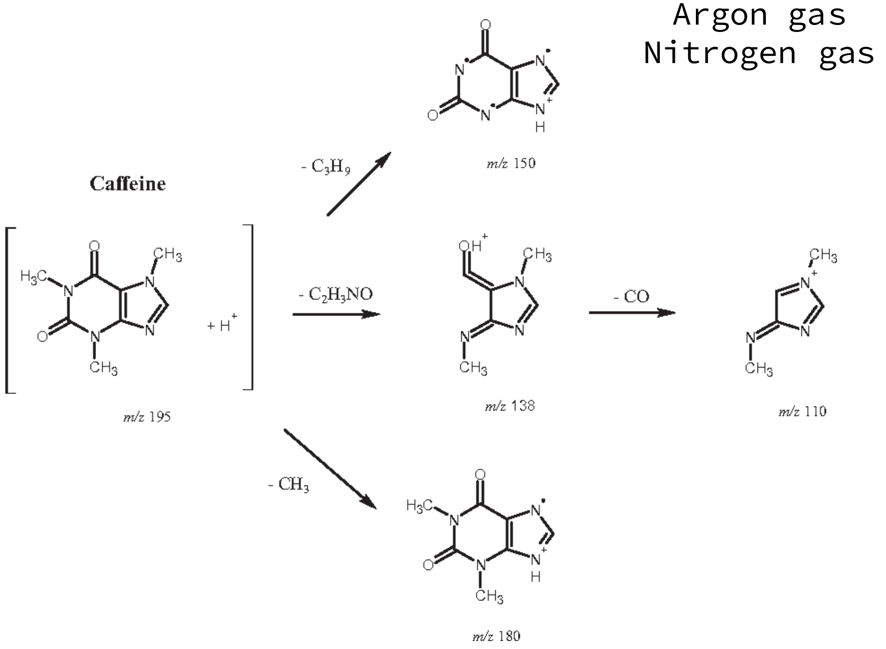


Q1 (MS^1)



Q2 (collision chamber)

Argon gas
Nitrogen gas



Pellegrini, Manuela et al. Rapid communications in mass spectrometry (2007).

Metabolomics data acquisition

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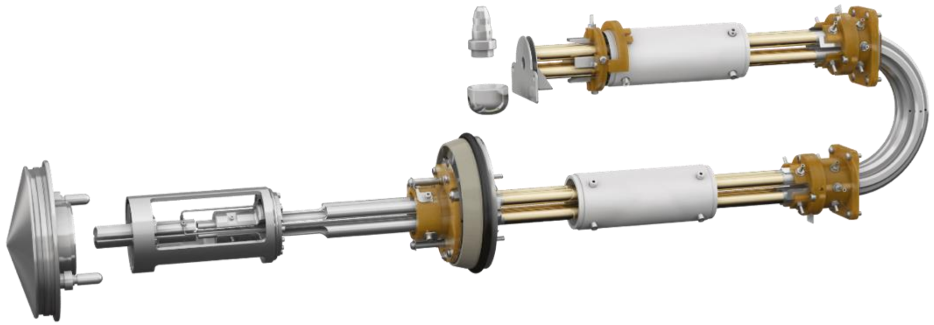
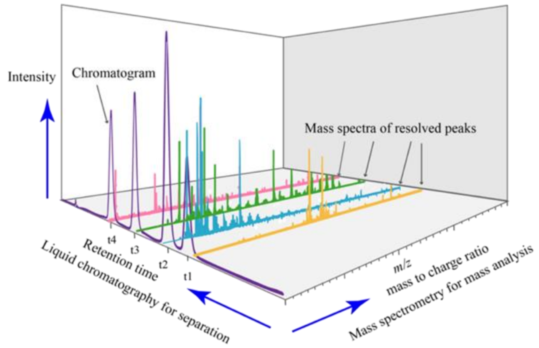


Q1(MS¹)



Q2(collision chamber)

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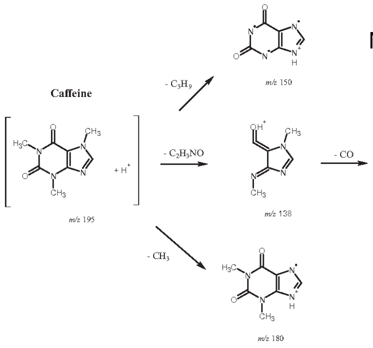
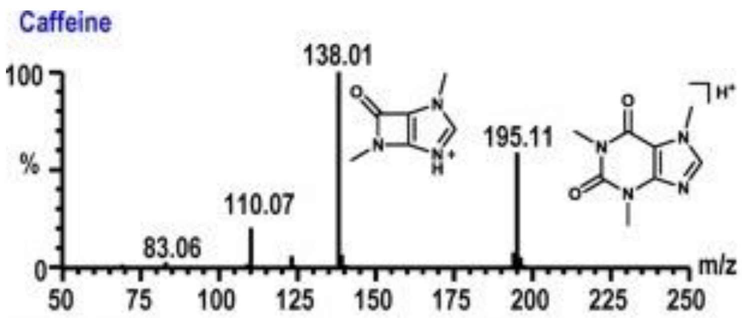


Q3(MS²)



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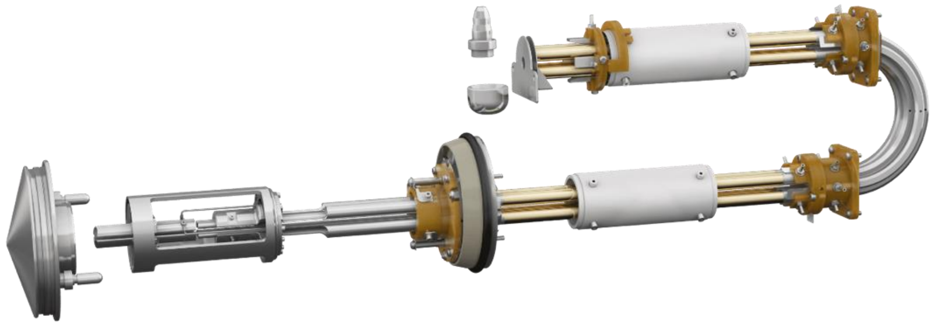
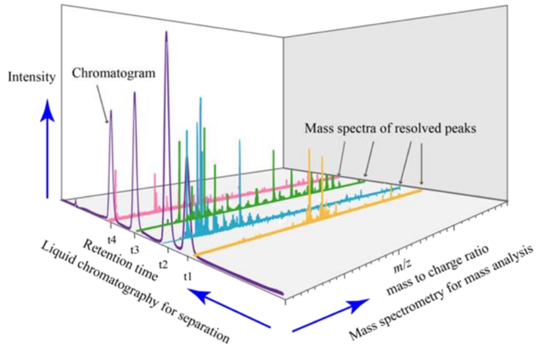
Metabolomics data acquisition

Liquid chromatography-Mass spectrometry (LC-MS)



Q1 (MS^1)



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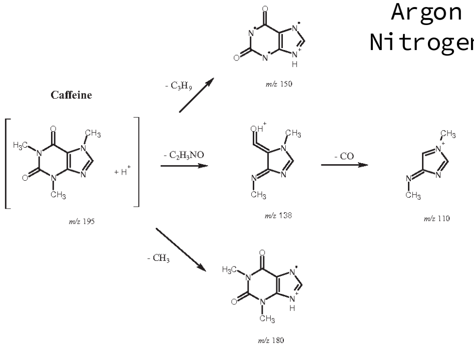
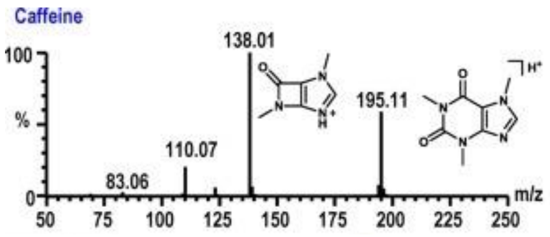


Argon gas
Nitrogen gas

Q3 (MS^2)



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Metabolomics data acquisition

Liquid chromatography-Mass spectrometry (LC-MS)

Data Dependent Acquisition (DDA)
Ions are selected based on intensity.
The MS makes real-time decisions during the run: MS^1 scan → Selection of Top K most intense ions

Pros
High-quality MS/MS spectra

Cons
Biased toward abundant metabolites

Metabolomics data acquisition

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Pros	Cons
High-quality MS/MS spectra	Biased toward abundant metabolites

Data Independent Acquisition (DIA)
It fragments all ions within predefined m/z windows.

Pros	Cons
Much more comprehensive coverage	MS/MS spectra are complex and overlapping

Metabolomics data acquisition

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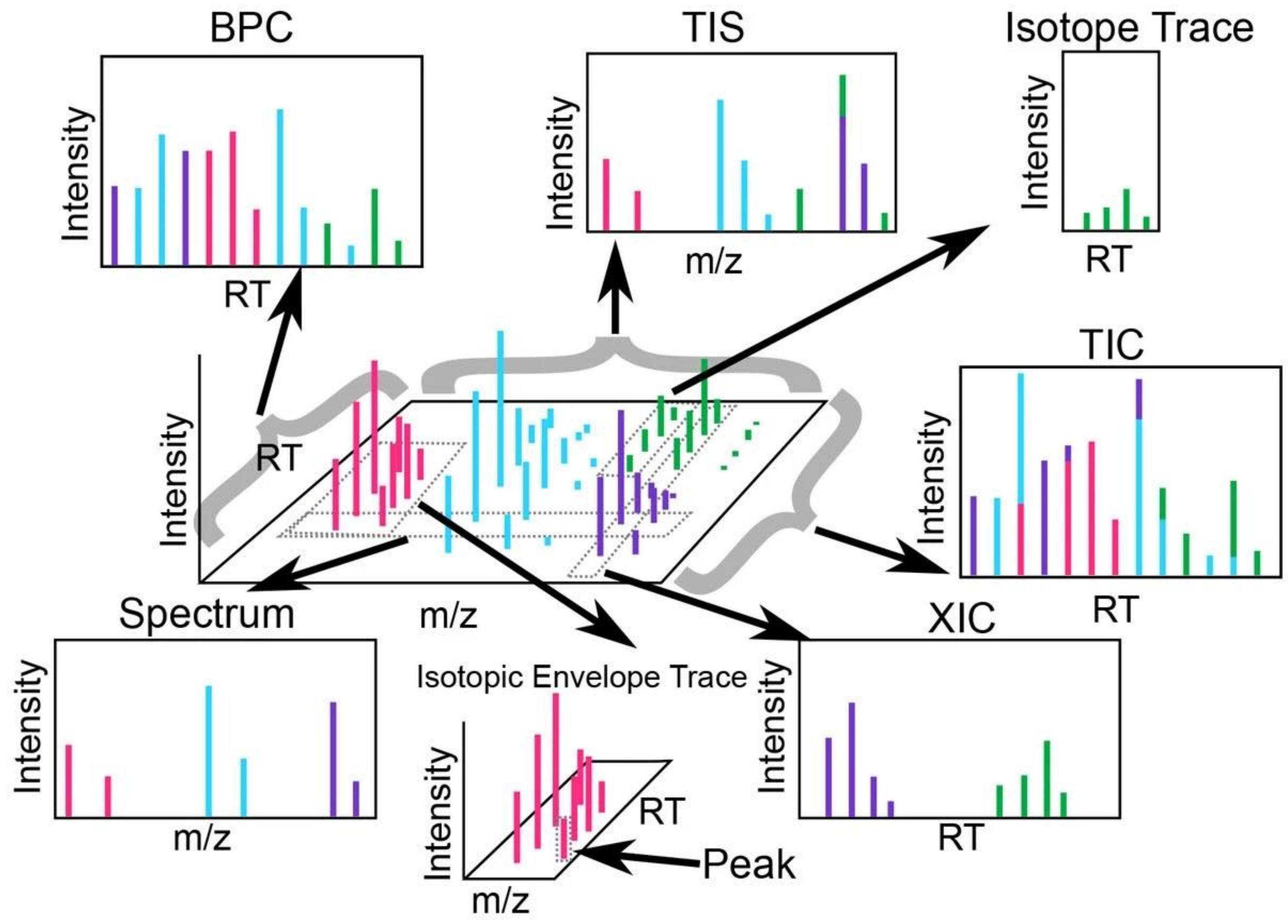
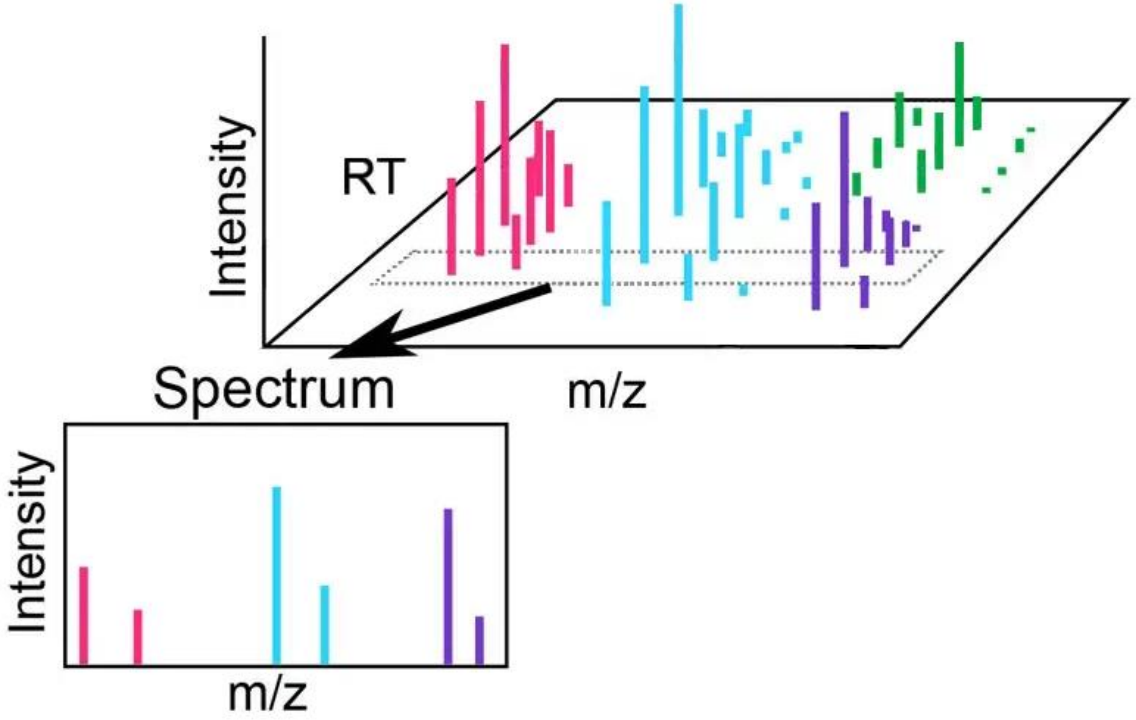
LC-MS data

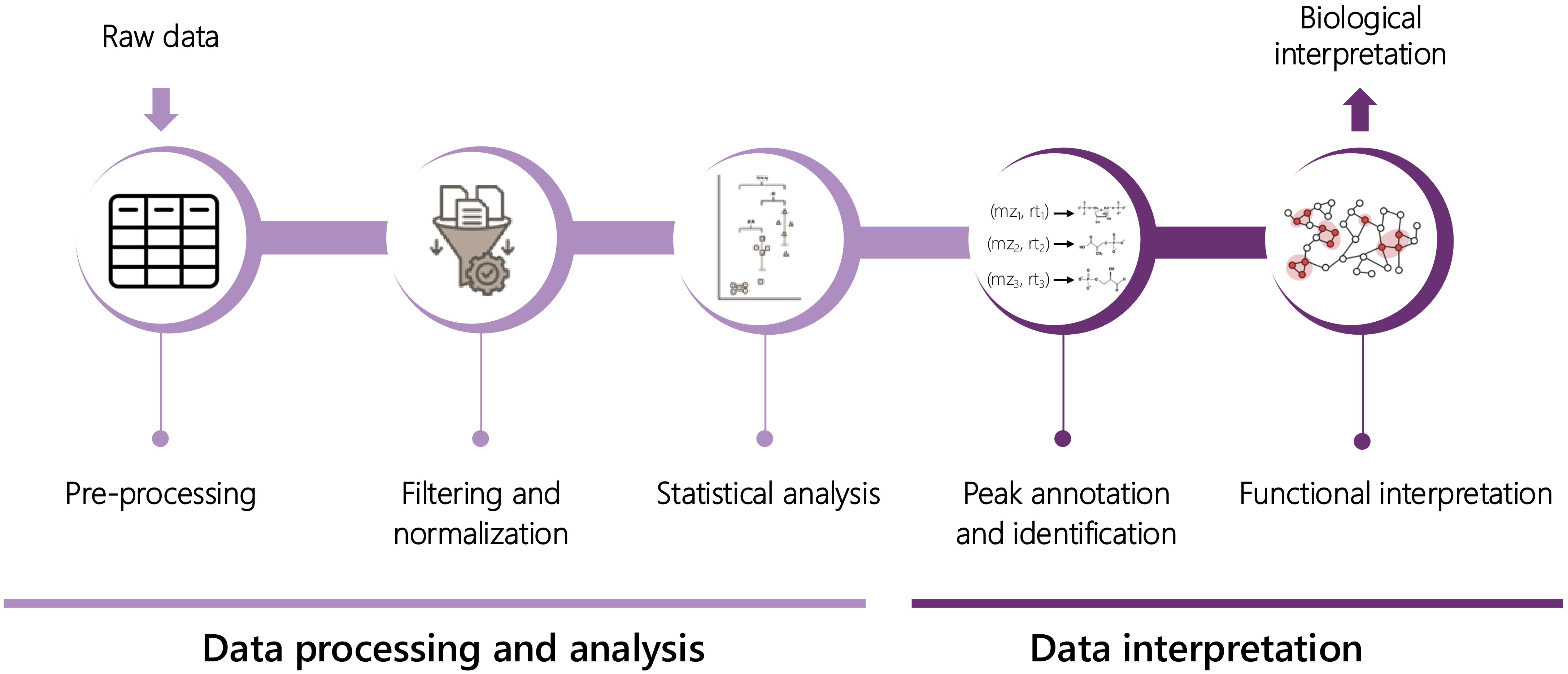


BPC
Intensity
RT
TIS
Intensity
m/z
Isotope Trace
Intensity
RT
TIC
Intensity
RT
Intensity
RT
m/z
Spectrum
Intensity
m/z
Isotopic Envelope Trace
Intensity
RT
m/z
Peak
XIC
Intensity
RT

LC-MS data

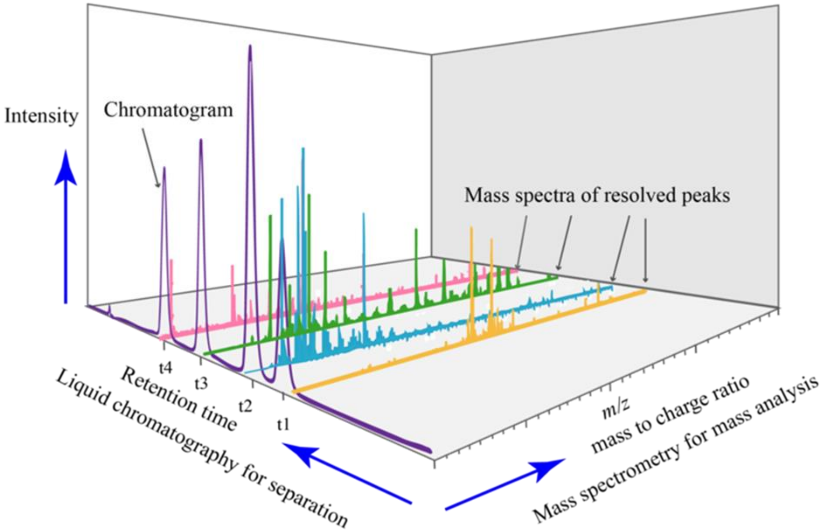


Computational metabolomics pipeline



Pre-processing

The goal is to convert **raw LC-MS data** into a **table of n peaks** defined by m/z, rt and an intensity value for each sample k.



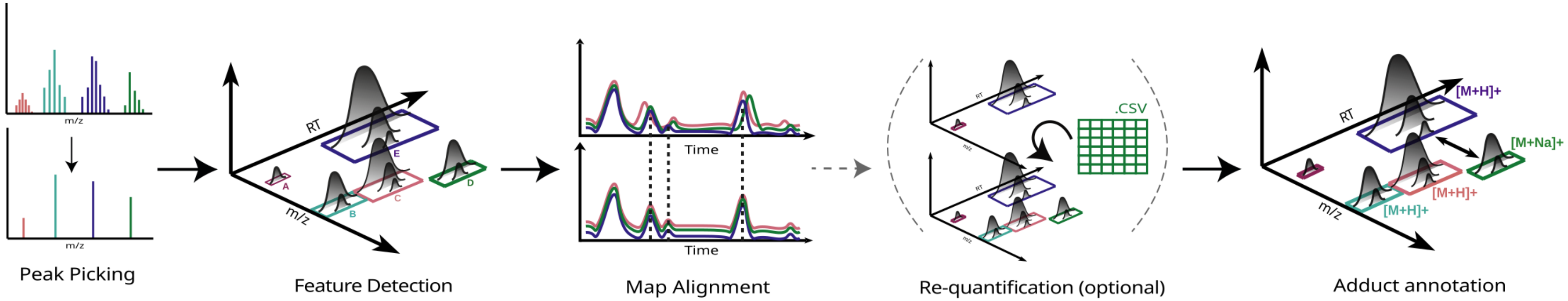
— Pre-processing →

m/z	**rt**	$\mathbf{I_1}$	**…**	$\mathbf{I_k}$
mz_1	rt_1	I_{11}	…	I_{1k}
…	…	…	…	…
mz_n	rt_n	I_{n1}	…	I_{nk}

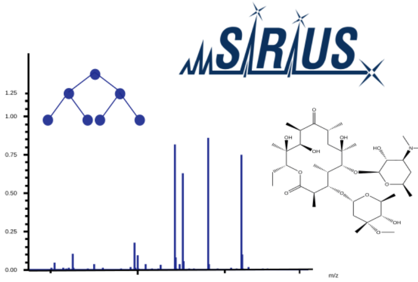
Pre-processing

Metaboigniter pipeline

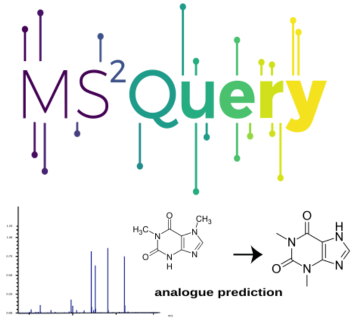
Pre-Processing



Identification (optional)



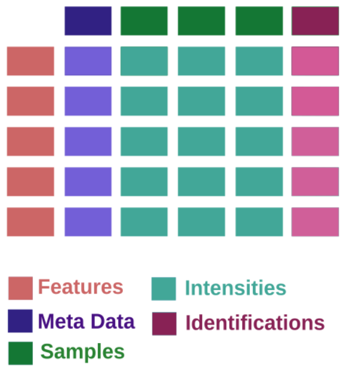
Predict metabolite formulas and structures with SIRIUS and CSI:FingerID.



Detect analogue molecules from your MS2 spectra with the machine learning tool MS2Query.

Feature Matrix

Feature Linking



Pre-processing

Centroiding

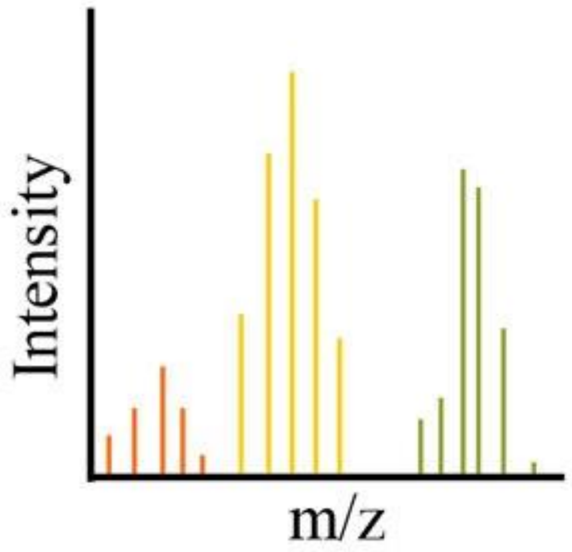
- The MS has finite resolving power → ions are detected as profile peaks around their true m/z rather than as a single exact m/z
- First step: centroiding estimates the most probable peak center (Continuous mass spectra → series of discrete points)
- Reduction of amount of data without much loss of information.
- The next processing steps require centroided data.
- Many manufacturers already apply centroiding of the profile data (during acquisition or immediately after).

Pre-processing

Centroiding with OpenMS PeakPickerHiRes

Profile mode

MS1 spectrum (scan at RT = 5.2 min)



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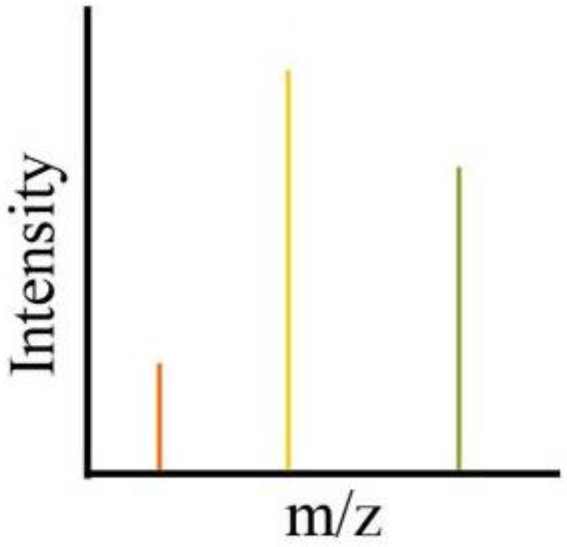

An Automated Pipeline for High-Throughput Label-Free Quantitative Proteomics

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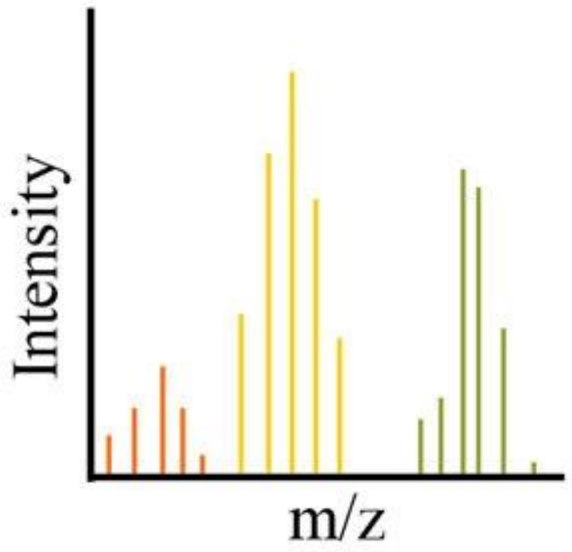

1. Detect regions of neighbouring data points that could be a peak. At least 3 points that satisfy:
   - Signal-to-noise filter (user-defined)
   - Evenly spaced across m/z
   - Peak shape: intensities first increase and then decrease

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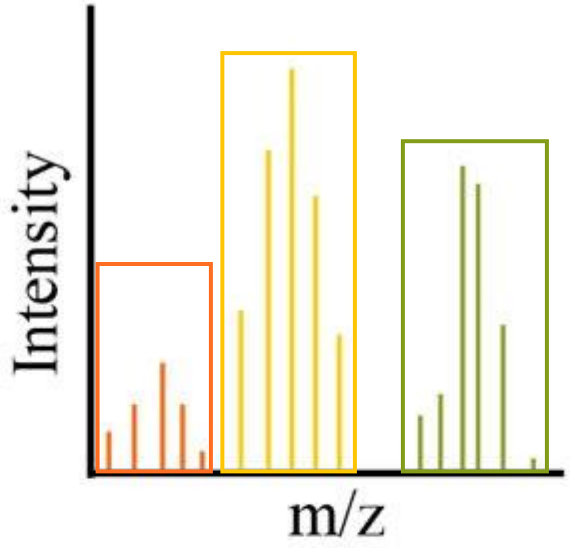

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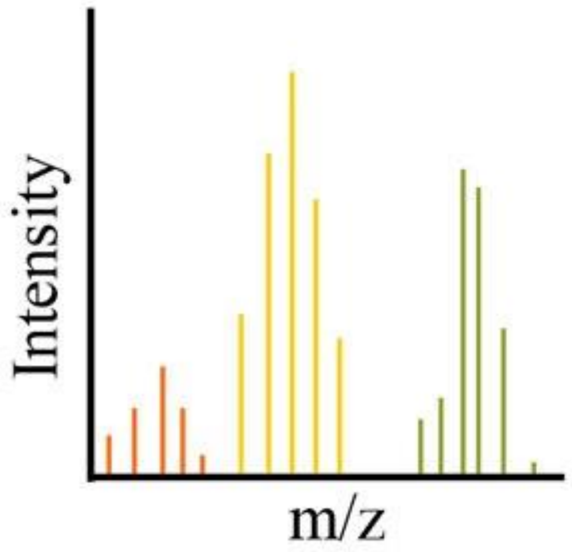

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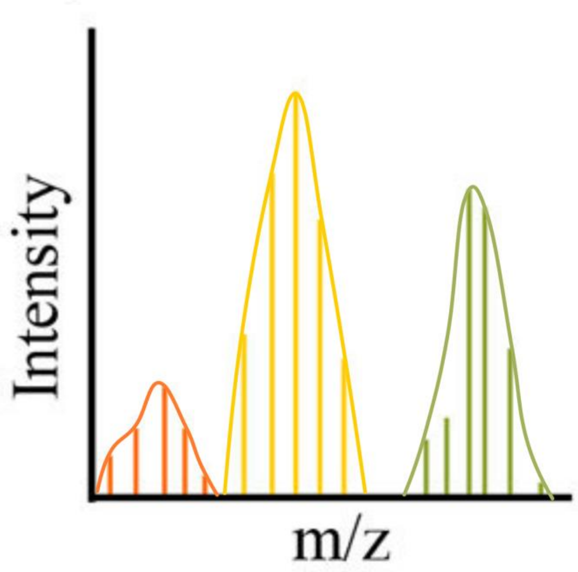

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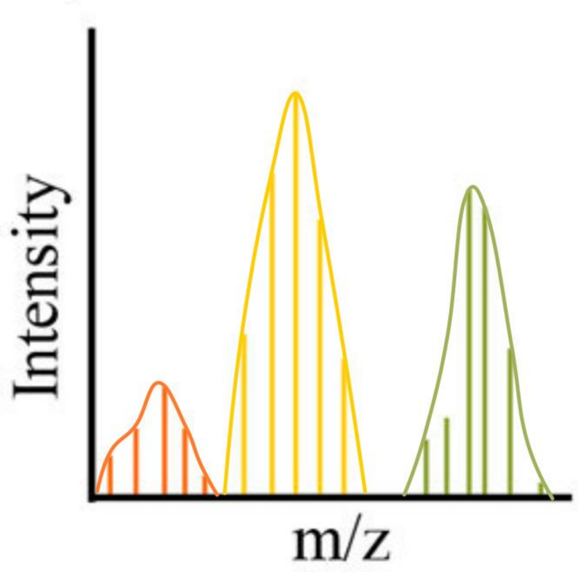


3. Find the peak region's apex (centroid):
   - Derivative of the spline: f (x) = I' (x) = 0
   - Bisection method
   - Peak region:

     m/z = [100.001, 100.005]

     f (100.001) = 4 and f (100.005) = −2

     m = (100.001+100.005)/2 = 100.003

     f (100.003) = +1 – Apex in [100.003, 100.005]

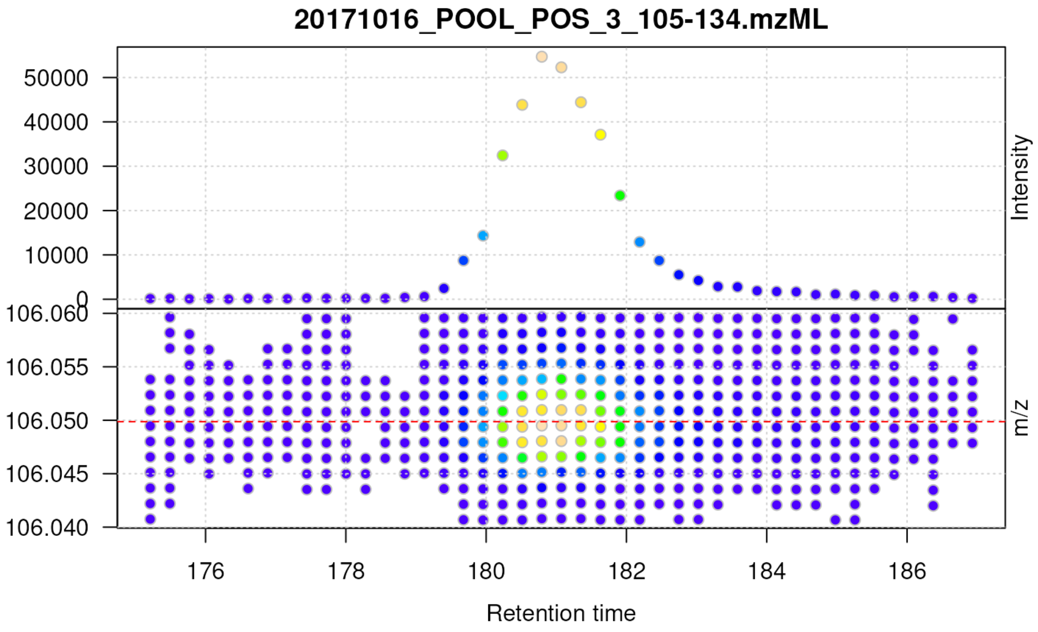
↓

Centroid m/z -- I (m/z) = centroid intensity

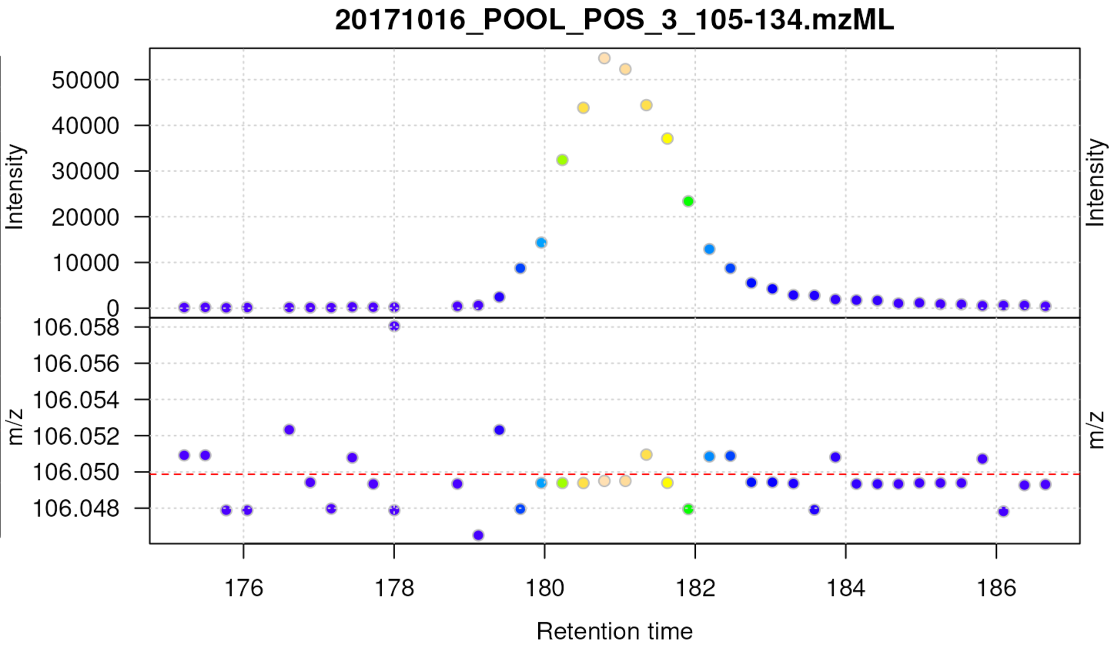
Pre-processing

Peak detection

Profile data



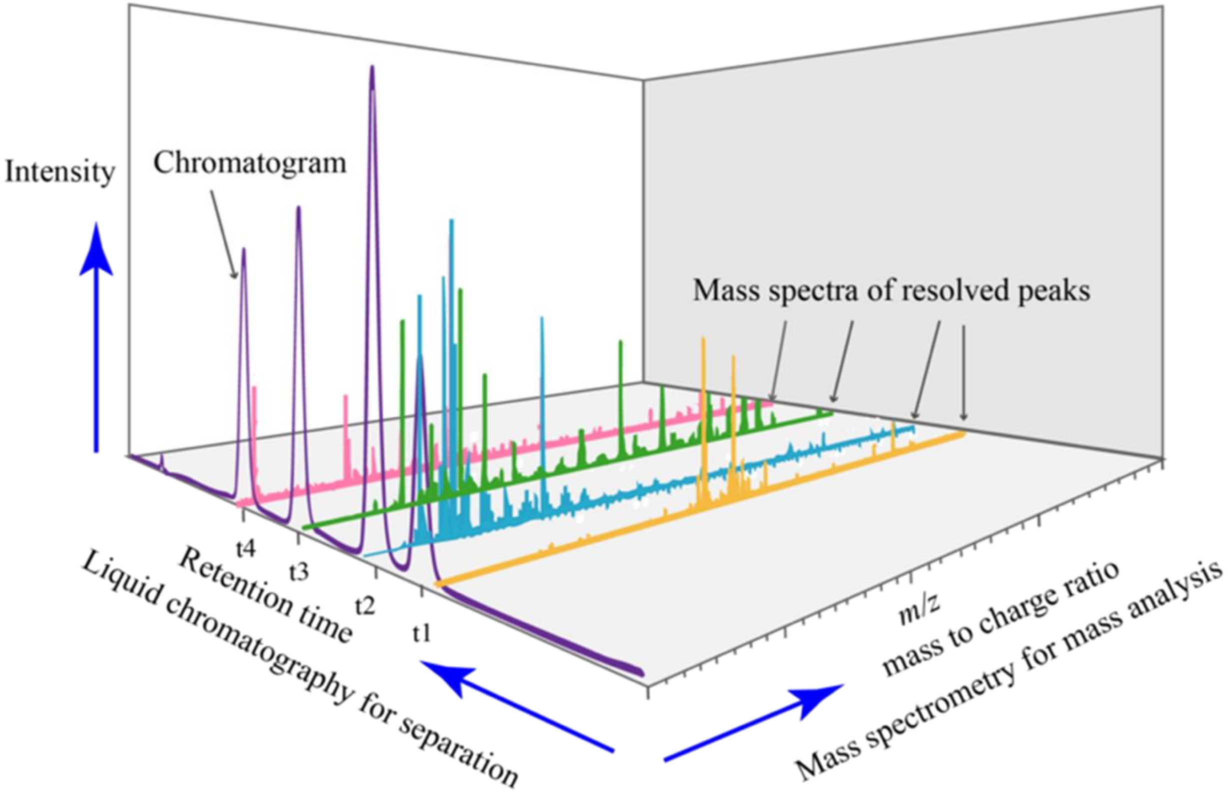
Centroid data



The **goal** is to identify the **chromatographic peaks** (peaks across the retention time axis) and return a **peak feature** defined by m/z, rt and intensity per sample.

Pre-processing

Peak detection with OpenMS FeatureFinderMetabo



1. Mass trace detection
2. Feature assembly

Kenar, E et al. (2014). Automated label-free quantification of metabolites from liquid chromatography-mass spectrometry data. Molecular & cellular proteomics.

Pre-processing

Peak detection with OpenMS FeatureFinderMetabo

1. **Mass trace detection**

$\mathrm{P} = \{\mathrm{p_k}\}$ where $\mathrm{p_k} = \{\mathrm{rt_k}, \mathrm{m/z_k}, \mathrm{I_k}\}$

- Sort P by decreasing intensity
- Remove peaks with I_k< user-defined threshold

↓

Filtered list P'

Mass trace T ⟶ List of n peaks $p_k \in P'$ with similar m/z and occur in adjacent scans of an LC-MS run.

Kenar, E et al. (2014). Automated label-free quantification of metabolites from liquid chromatography-mass spectrometry data. Molecular & cellular proteomics.

Pre-processing – Peak detection

OpenMS FeatureFinderMetabo

1. Mass trace detection

Mass trace T ⟶ List of n peaks $\mathrm{p_k} \in \mathrm{P'}$ with similar m/z and occur in adjacent scans of an LC-MS run.

1. A peak $\mathrm{p_k}$ is considered the start of a mass trace T: $\mathrm{T} = \{\mathrm{p_k}\}$
2. Extend trace T in time (both forward and backward):
    - Peaks in neighbouring scans
    - With Similar m/z values (probabilistic)
    - Add peaks to trace T
3. Trace extension stops when no more peaks can be added
4. Used peaks are removed to avoid duplication

Kenar, E et al. (2014). Automated label-free quantification of metabolites from liquid chromatography-mass spectrometry data. Molecular & cellular proteomics.

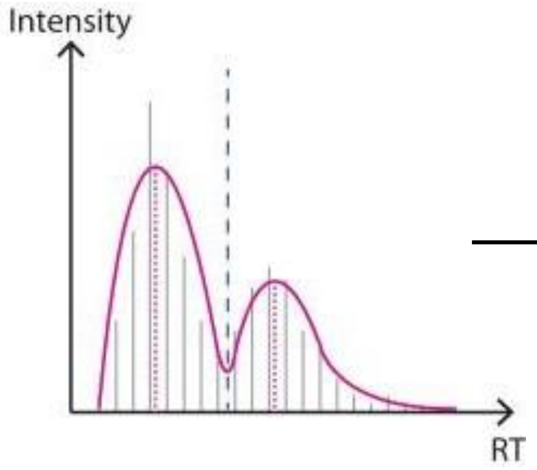
Pre-processing – Peak detection

OpenMS FeatureFinderMetabo

1. Mass trace detection

Two compounds can have:

- Almost identical m/z
- Overlap in RT



1. Smooth the signal with LOWESS
2. Find:
   - Local maxima (peaks)
   - Minima between maxima (separation points)
3. Each split becomes a final mass trace

Kenar, E et al. (2014). Automated label-free quantification of metabolites from liquid chromatography-mass spectrometry data. Molecular & cellular proteomics.

Pre-processing – Peak detection

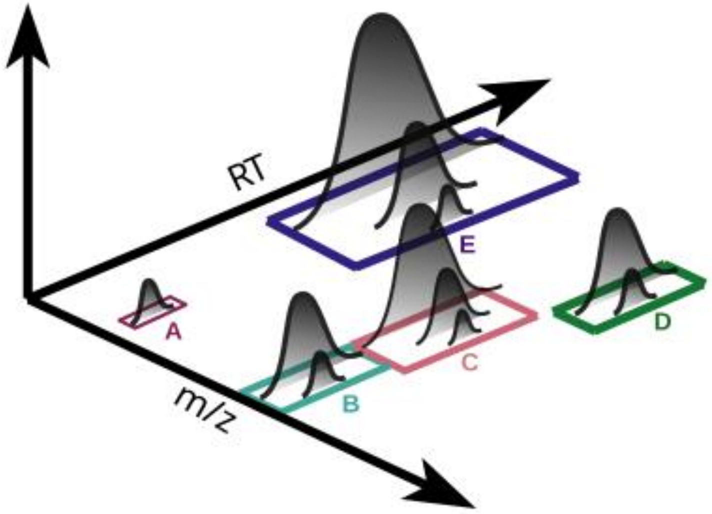
OpenMS FeatureFinderMetabo

2. Feature assembly

The goal is to find sets of isotopic mass traces that might originate from the same analyte.

A set is considered a valid isotope pattern if:

- Mass traces co-elute
- Correct m/z distances with respect to charge z
- Correct isotopic abundance ratios



Kenar, E et al. (2014). Automated label-free quantification of metabolites from liquid chromatography-mass spectrometry data. Molecular & cellular proteomics.

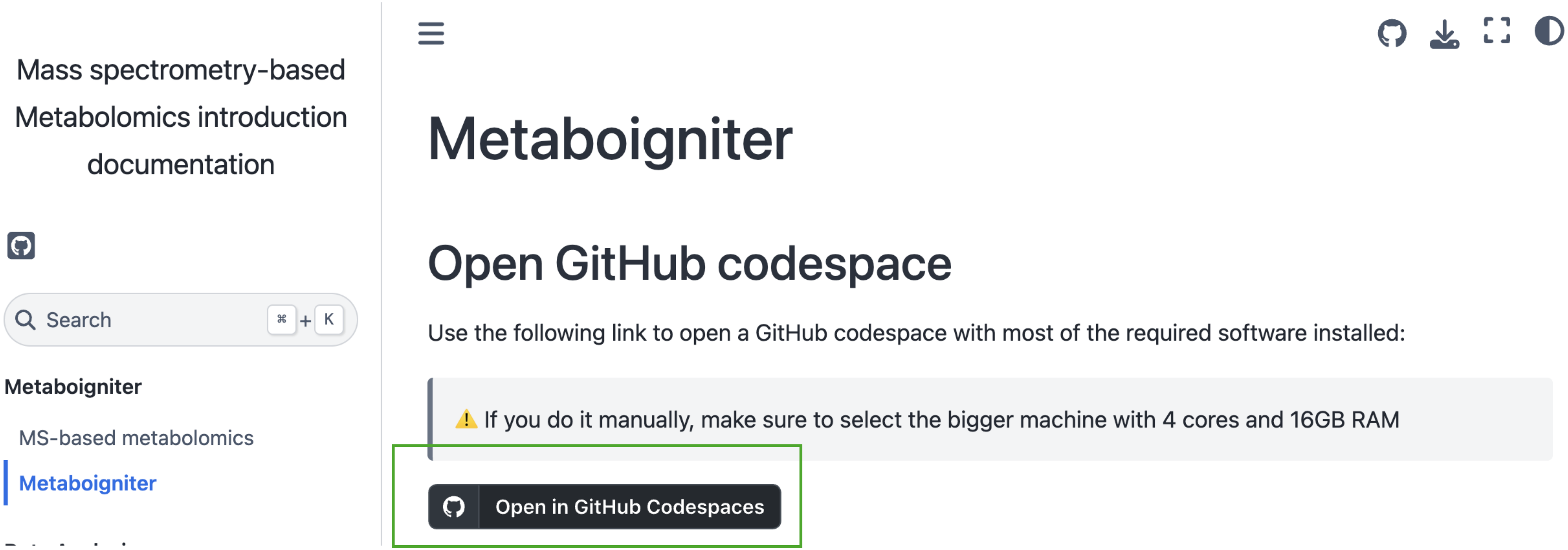
Pre-processing – Peak alignment and matching

The goal is to match features across different samples together:

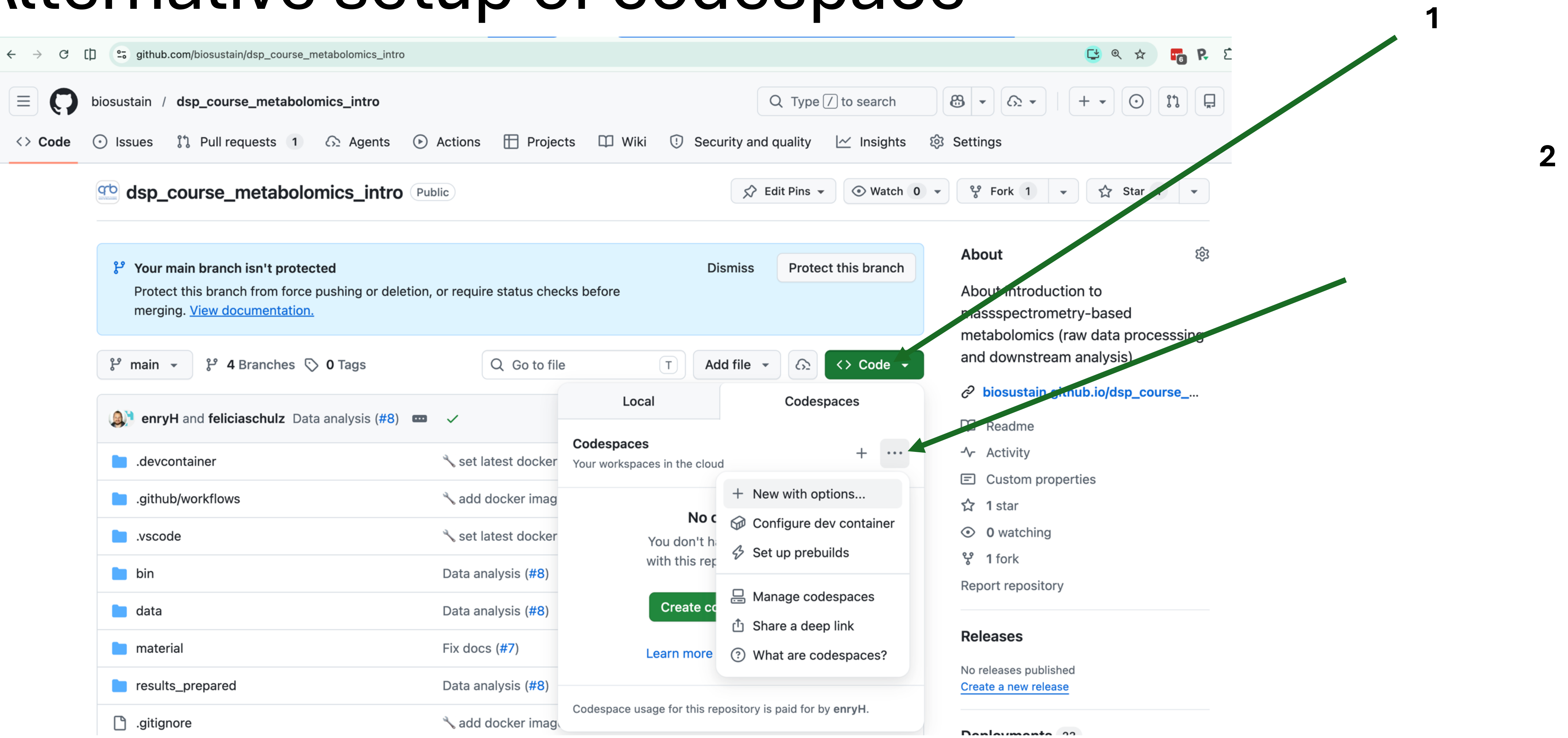
1. Peak alignment across the retention time axis
2. Grouping corresponding features in multiple runs

Pre-processing – Hands on

Click on: Open in GitHub Codespaces

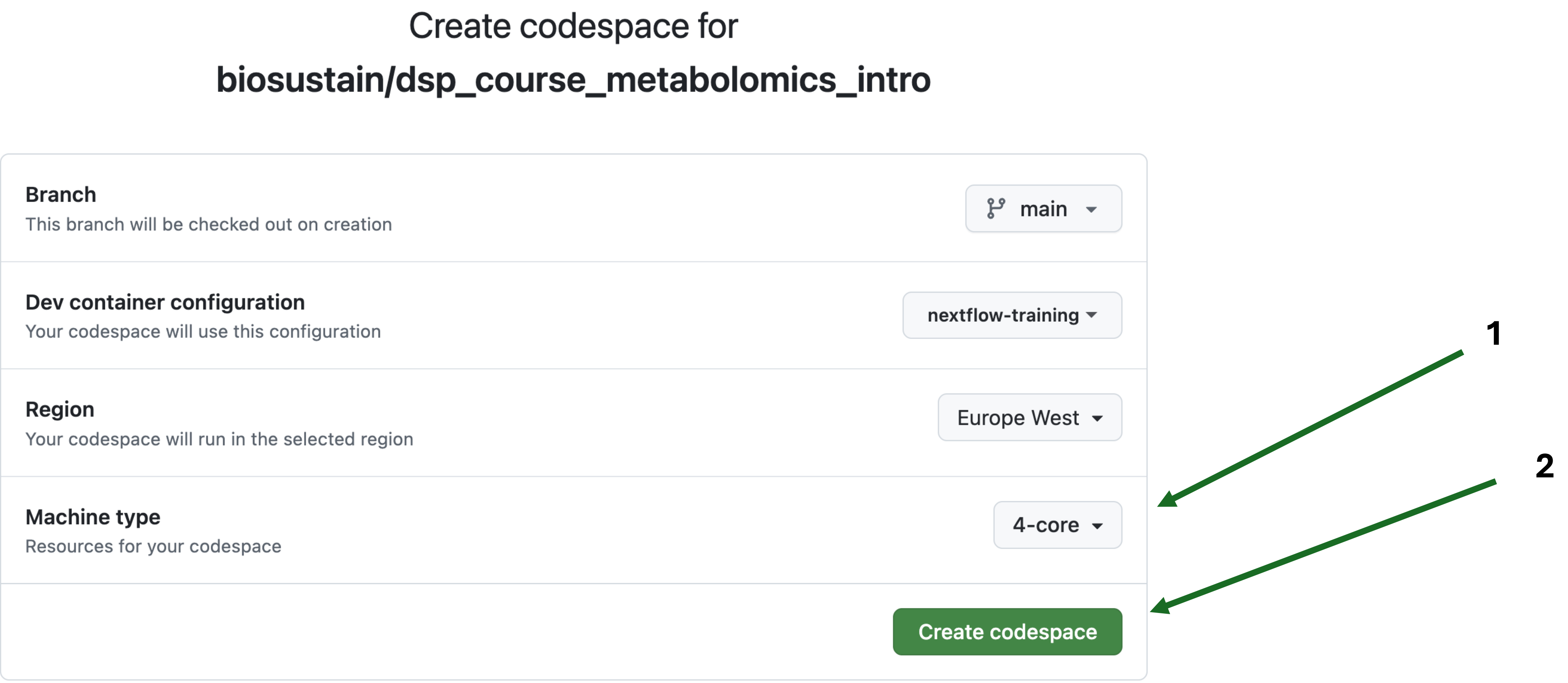


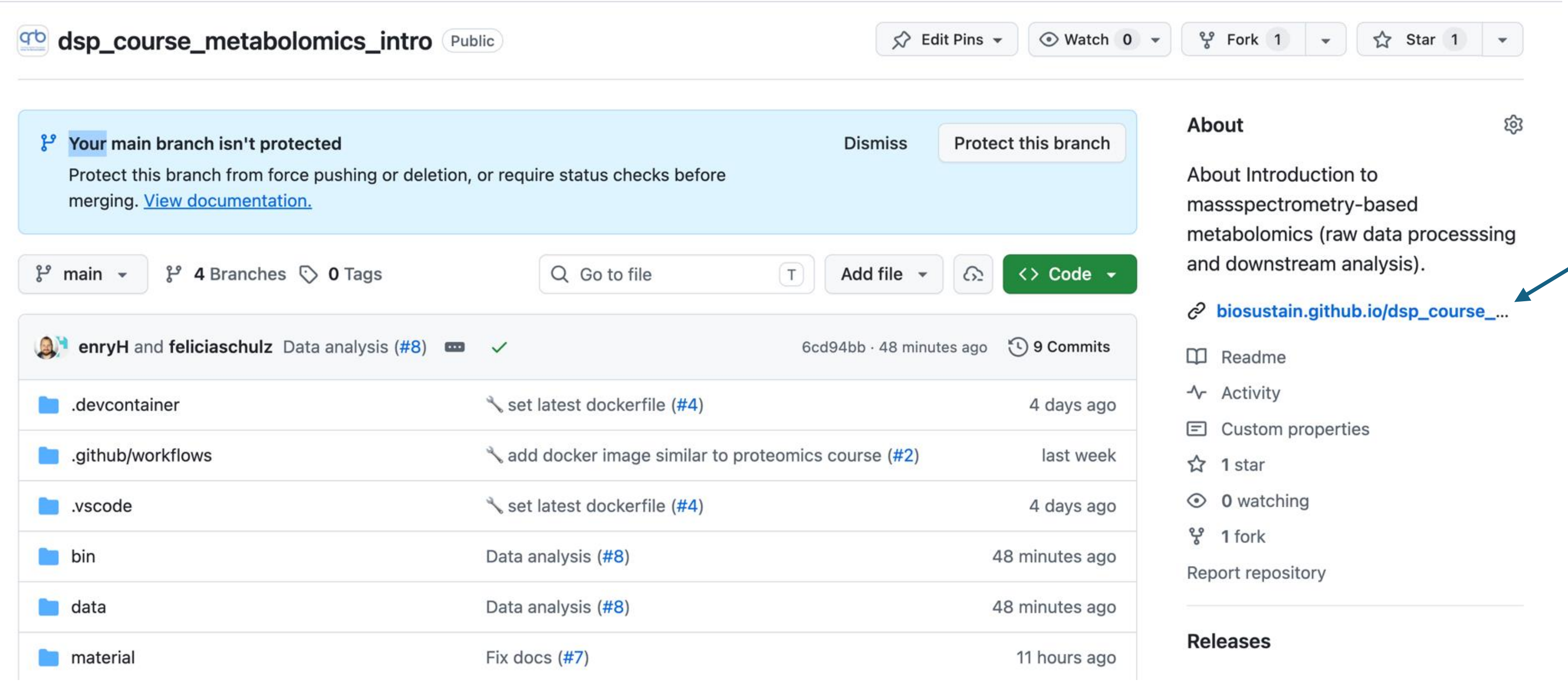
Alternative setup of codespace



Codespace with 4 cores (and 16GB of memory)

- Not Safari
- Ignore Errors (todo tree)





Go to website

Repo Overview

Feel free to ask questions at any point

enryH and feliciaschulz Data analysis (#8)		6cd94bb · 49 minutes ago · 9 Commits
.devcontainer	set latest dockerfile (#4)	4 days ago
.github/workflows	add docker image similar to proteomics course (#2)	last week
.vscode	set latest dockerfile (#4)	4 days ago
bin	Data analysis (#8)	49 minutes ago
data	Data analysis (#8)	49 minutes ago
material	Fix docs (#7)	11 hours ago
results_prepared	Data analysis (#8)	49 minutes ago
.gitignore	add docker image similar to proteomics course (#2)	last week
2_data_analysis.ipynb	Data analysis (#8)	49 minutes ago
2_data_analysis.py	Data analysis (#8)	49 minutes ago
README.md	Updated Agenda and Location June 2026 (#6)	4 days ago
conf.py	Data analysis (#8)	49 minutes ago
index.md	Data analysis (#8)	49 minutes ago
nextflow.config	set latest dockerfile (#4)	4 days ago
requirements.txt	Data analysis (#8)	49 minutes ago
requirements_docs.txt	🎉 initialize relatively bare bone repo for metabolomics cou...	3 months ago

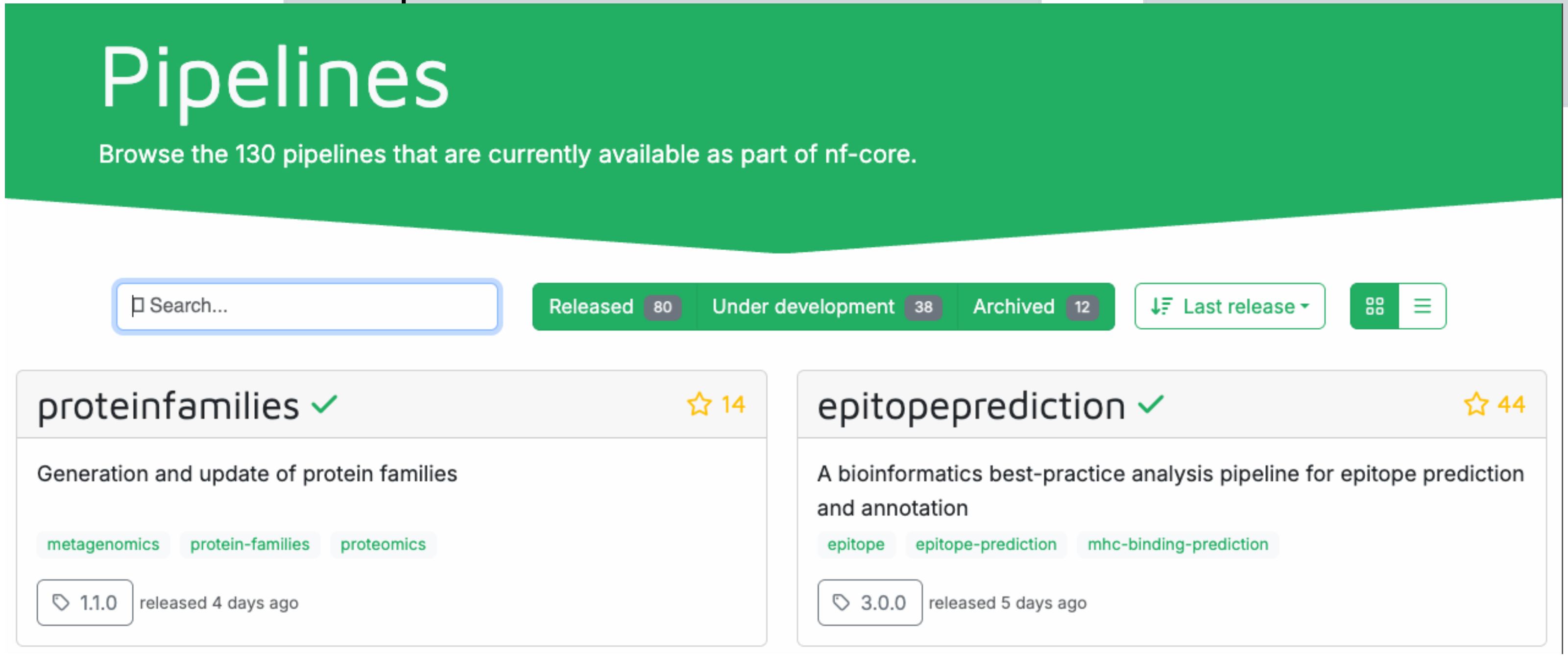
Nextflow and nf-core pipeline template

Nextflow is a workflow executor (analysis steps combined in an execution graph)

- Reproducibile and scalable

Nf-core is a collection of workflows maintained by nextflow and an open-source community

A global community effort to collect a curated set of open-source analysis pipelines built using Nextflow.

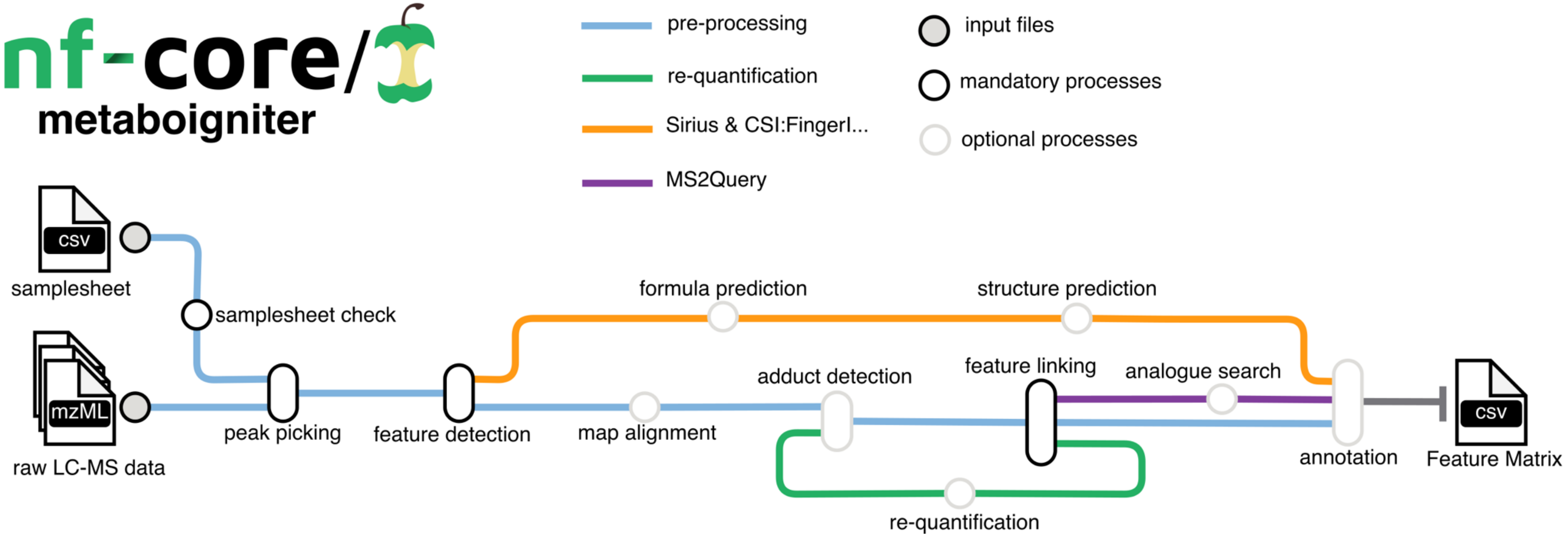


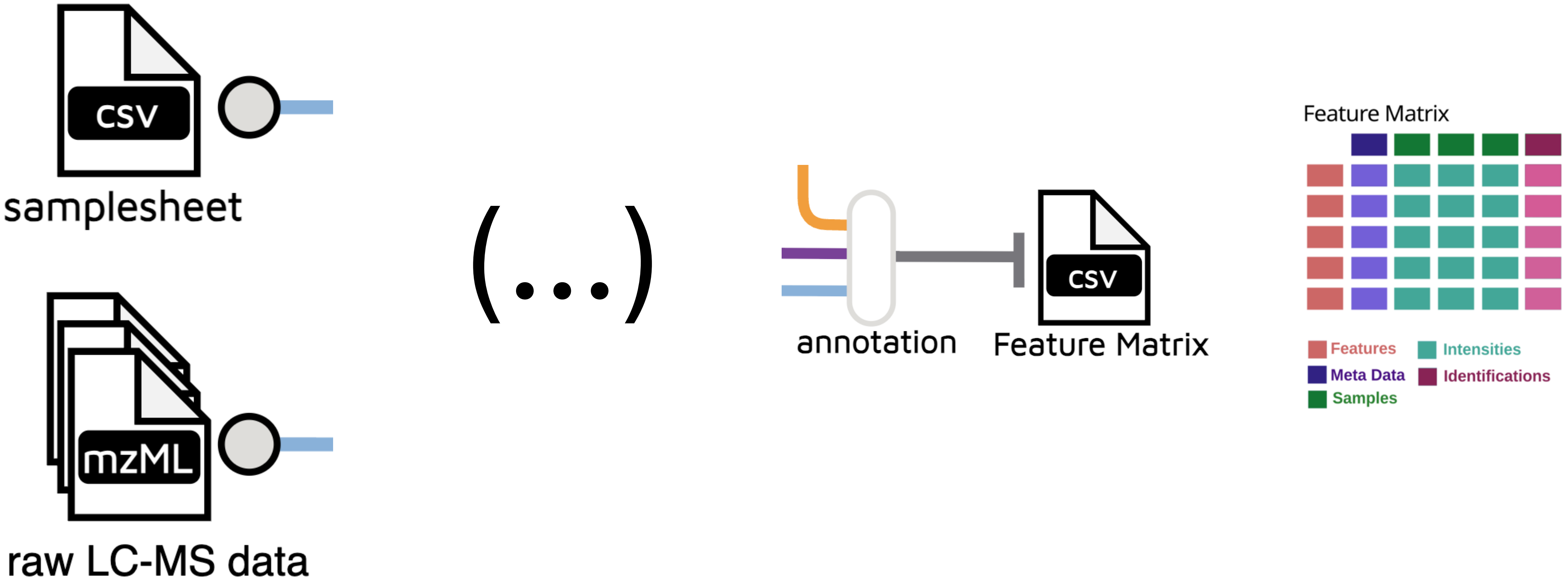
Works with

- Linux machines (on servers)
- on MacOS

INPUTS

Metaboigniter – Metro Map





CSV
samplesheet
mzML
raw LC-MS data
(...)
annotation
CSV
Feature Matrix
Feature Matrix
Features
Intensities
Meta Data
Identifications
Samples

Samplesheet

sample	level	type	msfile
MS_A_POS	MS1	normal	data/MTBLS8735/MS_A_POS.mzML
MS_B_POS	MS1	normal	data/MTBLS8735/MS_B_POS.mzML
MS_C_POS	MS1	normal	data/MTBLS8735/MS_C_POS.mzML
MS_D_POS	MS1	normal	data/MTBLS8735/MS_D_POS.mzML
MS_E_POS	MS1	normal	data/MTBLS8735/MS_E_POS.mzML
MS_F_POS	MS1	normal	data/MTBLS8735/MS_F_POS.mzML
MS_QC_POOL_1_POS	MS1	normal	data/MTBLS8735/MS_QC_POOL_1_POS.mzML
MS_QC_POOL_2_POS	MS1	normal	data/MTBLS8735/MS_QC_POOL_2_POS.mzML
MS_QC_POOL_3_POS	MS1	normal	data/MTBLS8735/MS_QC_POOL_3_POS.mzML
MS_QC_POOL_4_POS	MS1	normal	data/MTBLS8735/MS_QC_POOL_4_POS.mzML
MSMS_2_A_CE20_POS	MS2	normal	data/MTBLS8735/MSMS_2_A_CE20_POS.mzML
MSMS_2_A_CE30_POS	MS2	normal	data/MTBLS8735/MSMS_2_A_CE30_POS.mzML
MSMS_2_A_CES_POS	MS2	normal	data/MTBLS8735/MSMS_2_A_CES_POS.mzML
MSMS_2_E_CE20_POS	MS2	normal	data/MTBLS8735/MSMS_2_E_CE20_POS.mzML
MSMS_2_E_CE30_POS	MS2	normal	data/MTBLS8735/MSMS_2_E_CE30_POS.mzML
MSMS_2_E_CES_POS	MS2	normal	data/MTBLS8735/MSMS_2_E_CES_POS.mzML

Spectra files

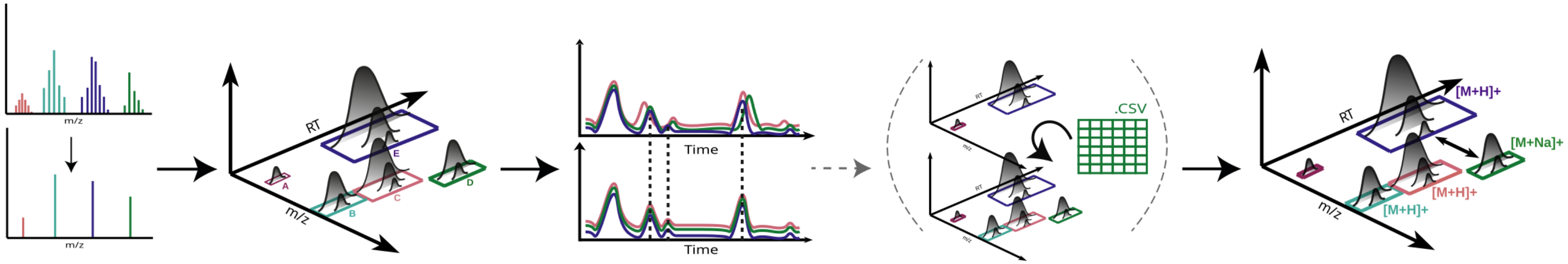
- Open standard, text based:
  - mzML files (indexed)
- Needs to be created from .d (bruker), .raw (Thermo), etc. vendor specific file formats
  - Conversion not (yet) integrated into workflow

One MS1 spectrum (DDA)


```xml
<spectrum id="controllerType=0 controllerNumber=1 scan=6" index="5" defaultArrayLength="736">
  <cvParam cvRef="MS" accession="MS:1000511" value="1" name="ms level" />
  <cvParam cvRef="MS" accession="MS:1000579" value="" name="MS1 spectrum" />
  <cvParam cvRef="MS" accession="MS:1000130" value="" name="positive scan" />
  <cvParam cvRef="MS" accession="MS:1000285" value="15248891" name="total ion current" />
  <cvParam cvRef="MS" accession="MS:1000127" value="" name="centroid spectrum" />
  <cvParam cvRef="MS" accession="MS:1000504" value="401.923461914063" name="base peak m/z" unitAccession="MS:1000040" unitName="m/z" unitCvRef="MS" />
  <cvParam cvRef="MS" accession="MS:1000505" value="2704116.25" name="base peak intensity" unitAccession="MS:1000131" unitName="number of detector counts" unitCvRef="MS" />
  <cvParam cvRef="MS" accession="MS:1000528" value="375.884124755859" name="lowest observed m/z" unitAccession="MS:1000040" unitName="m/z" unitCvRef="MS" />
  <cvParam cvRef="MS" accession="MS:1000527" value="1665.40612792969" name="highest observed m/z" unitAccession="MS:1000040" unitName="m/z" unitCvRef="MS" />
  <scanList count="1">
    <cvParam cvRef="MS" accession="MS:1000795" value="" name="no combination" />
    <scan instrumentConfigurationRef="IC1">
      <cvParam cvRef="MS" accession="MS:1000016" value="6.0323342" name="scan start time" unitAccession="UO:0000031" unitName="minute" unitCvRef="UO" />
      <cvParam cvRef="MS" accession="MS:1000512" value="FTMS + p NSI Full ms [375.0000-1800.0000]" name="filter string" />
      <cvParam cvRef="MS" accession="MS:1000927" value="50" name="ion injection time" unitAccession="UO:0000028" unitName="millisecond" unitCvRef="UO" />
      <scanWindowList count="1">
        <scanWindow>
          <cvParam cvRef="MS" accession="MS:1000501" value="375" name="scan window lower limit" unitAccession="MS:1000040" unitName="m/z" unitCvRef="MS" />
          <cvParam cvRef="MS" accession="MS:1000500" value="1800" name="scan window upper limit" unitAccession="MS:1000040" unitName="m/z" unitCvRef="MS" />
        </scanWindow>
      </scanWindowList>
    </scan>
  </scanList>
  <binaryDataArrayList count="2">
    <binaryDataArray encodedLength="3152">
      <cvParam cvRef="MS" accession="MS:1000514" value="" name="m/z array" unitAccession="MS:1000040" unitName="m/z" unitCvRef="MS" />
      <cvParam cvRef="MS" accession="MS:1000523" value="" name="64-bit float" />
      <cvParam cvRef="MS" accession="MS:1000574" value="" name="zlib compression" />
      <binary>eJwt03lYVXUawPHjVumjGA1lG5dTGtMi3MYUG32496hJOiNX3EvlcjLNsEDNckGWIyoGxaIwiU3AsUWcZ2TJEnEm5Wimg48C466j15M12ZRgU1paCTPP+/399Xne9fceFk3TzIeylxna//0Vjck5ovM26h1oDLd
    </binaryDataArray>
    <binaryDataArray encodedLength="4356">
      <cvParam cvRef="MS" accession="MS:1000515" value="" name="intensity array" unitAccession="MS:1000131" unitName="number of counts" unitCvRef="MS" />
      <cvParam cvRef="MS" accession="MS:1000523" value="" name="64-bit float" />
      <cvParam cvRef="MS" accession="MS:1000574" value="" name="zlib compression" />
      <binary>eJwtWHlcT+kXviLFSI2MTJi5ZWIaTTGDsl+EkT3KVtypUVl+dhr5TG4qo7Jk8FPWm32pjC1kuxUlEu0JuSVDpozK0pdkPj3P/PV+3vu+73nPec45zznvFQRBuRdyQxIEQWzvf6ppFBz63sIYceZq06jafJHbNGr
    </binaryDataArray>
  </binaryDataArrayList>
</spectrum>
```

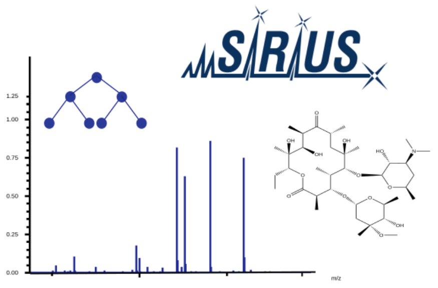
Metaboigniter - Overview

Pre-Processing

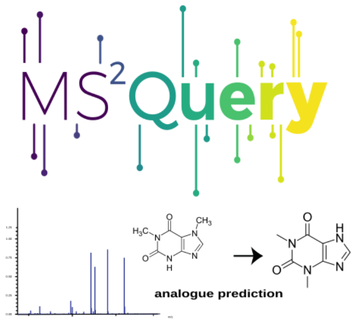


Peak Picking → Feature Detection → Map Alignment → Re-quantification (optional) → Adduct annotation

Identification (optional)



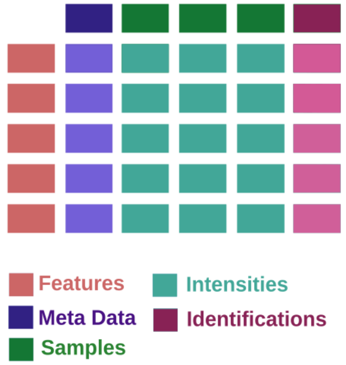
Predict metabolite formulas and structures with SIRIUS and CSI:FingerID.



Detect analogue molecules from your MS2 spectra with the machine learning tool MS2Query.

Feature Matrix

Feature Linking



pyOpenMS

OpenMS

Data analysis - Filtering and normalization

Input

Peak	m/z	rt	I_1	…	I_k
P_1	m/z_1	rt_1	I_{11}	…	I_{1k}
…	…	…	…	…	…
P_n	m/z_n	rt_n	I_{n1}	…	I_{nk}

Data analysis - Filtering and normalization

Input

Peak	m/z	rt	I_1	…	I_k
P_1	m/z_1	rt_1	I_{11}	…	I_{1k}
…	…	…	…	…	…
P_n	m/z_n	rt_n	I_{n1}	…	I_{nk}

Data analysis - Filtering and normalization

Input

Peak	m/z	rt	I_1	…	I_k
P_1	m/z_1	rt_1	I_{11}	…	I_{1k}
…	…	…	…	…	…
P_n	m/z_n	rt_n	I_{n1}	…	I_{nk}

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Peak	m/z	rt	I_1	…	I_k
P_1	m/z_1	rt_1	I_{11}	…	I_{1k}
…	…	…	…	…	…
P_n	m/z_n	rt_n	I_{n1}	…	I_{nk}

Data analysis - Filtering and normalization

Input

Peak	m/z	rt	I_1	…	I_k
P_1	m/z_1	rt_1	I_{11}	…	I_{1k}
…	…	…	…	…	…
P_n	m/z_n	rt_n	I_{n1}	…	I_{nk}

Data analysis - Filtering and normalization

Input

Peak	m/z	rt	I_1	...	I_k
P_1	m/z_1	rt_1	I_{11}	...	I_{1k}
...	...	...	...	...	...
P_n	m/z_n	rt_n	I_{n1}	...	I_{nk}

Missingness due to errors or real absence of the molecule

Data analysis - Filtering and normalization

Filtering using the 80% rules

Peak	m/z	rt	missigness (%)	I_1	…	I_k
P_1	m/z_1	rt_1	5%	I_{11}	…	I_{1k}
P_2	m/z_2	rt_2	8%	I_{21}	…	I_{2k}
P_3	m/z_3	rt_3	22%	I_{31}	…	I_{3k}
…	…	…	…	…	…	…
P_n	m/z_n	rt_n	42%	I_{n1}	…	I_{nk}

Data analysis - Filtering and normalization

Filtering using the 80% rules

Peak	m/z	rt	missigness (%)	I_1	…	I_k
P_1	m/z_1	rt_1	5%	I_{11}	…	I_{1k}
P_2	m/z_2	rt_2	8%	I_{21}	…	I_{2k}
~~P_3~~	~~m/z_3~~	~~rt_3~~	~~22%~~	~~I_{31}~~	~~…~~	~~I_{3k}~~
…	…	…	…	…	…	…
~~P_n~~	~~m/z_n~~	~~rt_n~~	~~42%~~	~~I_{n1}~~	~~…~~	~~I_{nk}~~

Data analysis - Statistical Analysis

Statistical analysis

MTBLS8735 - subset

Serum from

- three cardiovascular disease (CVD)
- Three healthy controls (CTR)
- Four quality control (QC) samples – pooled serum

https://rformassspectrometry.github.io/Metabonaut/articles/dataset-investigation.html

Data analysis – Hands on

Metabolomics Interpretation

Peak annotation vs peak identification

Identification levels according to the Chemical Analysis Working Group of the Metabolomics Standards Initiative:

Level 1 – Identified compounds require comparison of at least two independent properties (e.g., retention time and MS/MS spectra) with those of an authentic standard analysed under identical conditions.

Metabolomics Interpretation

Peak annotation vs peak identification

Identification levels according to the Chemical Analysis Working Group of the Metabolomics Standards Initiative:

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Level 2 – Putatively annotated compounds based on similarity to MS/MS data in commercial or public databases (reference standards are not available).

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Level 3 – Putatively characterized compound class.

Level 4 – Unknown feature.

Metabolomics Interpretation

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Level 4 – Unknown feature.

LC-MS[1]

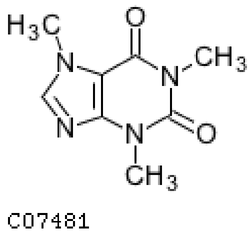
Metabolomics Interpretation

Peak annotation using LC-MS[1]

Peak annotation is a major bottleneck and direct matching of masses is not enough:

1

A single metabolite produces multiple peaks due to adducts and fragments.



Caffeine
Exact mass = 194.0804

— LC-MS + →

$[M+H]^+$

$[M+Na]^+$

$[M+K]^+$

$[M+NH_4]^+$

Metabolomics Interpretation

Peak annotation using LC-MS[1]

Peak annotation is a major bottleneck and direct matching of masses is not enough:

2

DB matching results in hundreds of potential candidates.

m/z = 195.0877 → hmdb The Human Metabolome Database →

Positive mode
10 different adducts

635 Results:
Caffeine $[M+H]^+$ - mass = 194.0804
Enprofylline $[M+H]^+$ - mass = 194.0804
Luminol $[M+NH_4]$ – mass = 177.0538
…

Metabolomics Interpretation

Peak annotation using LC-MS[1]

Peak annotation is a major bottleneck and direct matching of masses is not enough:

1. A single metabolite produces multiple peaks due to adducts and fragments.
2. DB matching results in hundreds of potential candidates.
3. The metabolome is not completely annotated.

⇩

Computational algorithms that use m/z, rt and intensity to return putative annotations.

Metabolomics Interpretation

Peak annotation using LC-MS[1]

Peak annotation is a major bottleneck and direct matching of masses is not enough:

1. A single metabolite produces multiple peaks due to adducts and fragments.
2. DB matching results in hundreds of potential candidates.
3. The metabolome is not completely annotated.

Computational algorithms ⚠ Low confidence level putative annotations.

Metabolomics Interpretation

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Identification levels according to the Chemical Analysis Working Group of the **Metabolomics Standards Initiative**:

Level 1 – Identified compounds require comparison of at least two independent properties (e.g., retention time and MS/MS spectra) with those of an authentic standard analysed under identical conditions.

Level 2 – Putatively annotated compounds based on similarity to MS/MS data in commercial or public databases (reference standards are not available).

LC-MS^2

Level 3 – Putatively characterized compound class.

Level 4 – Unknown feature.

Metabolomics Interpretation

Peak annotation using LC-MS^2 spectra

1. Spectral library matching

2. Fragmentation tree methods

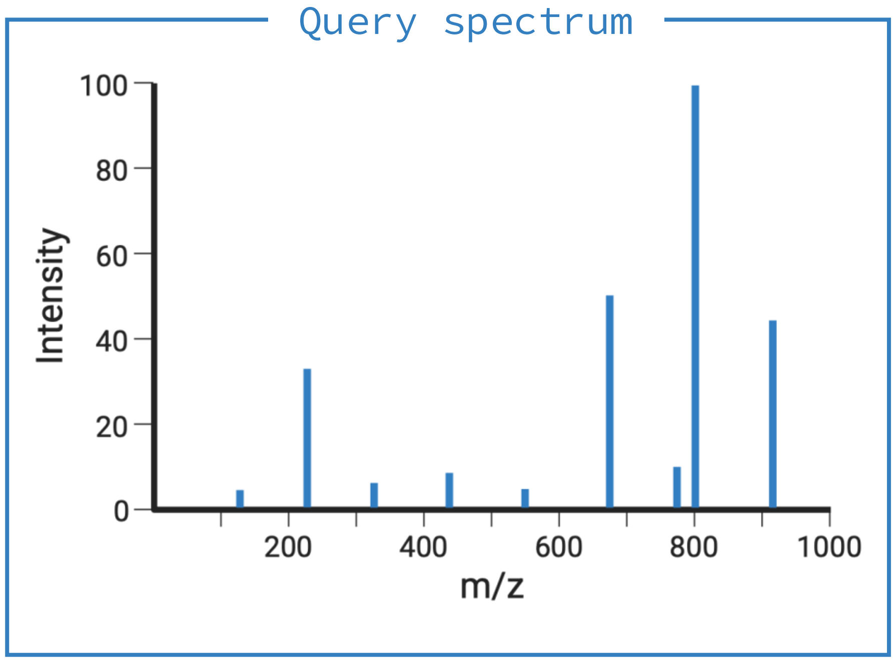
3. In-silico fragmentation

Add definitions, mention that in-silico spectra can then be used for spectral library matching, and that we focus on spectral library matching, maybe add here limitations of spectral library matching and then mention in-silico and analogue search

Metabolomics Interpretation

Peak annotation using LC-MS^2 spectra

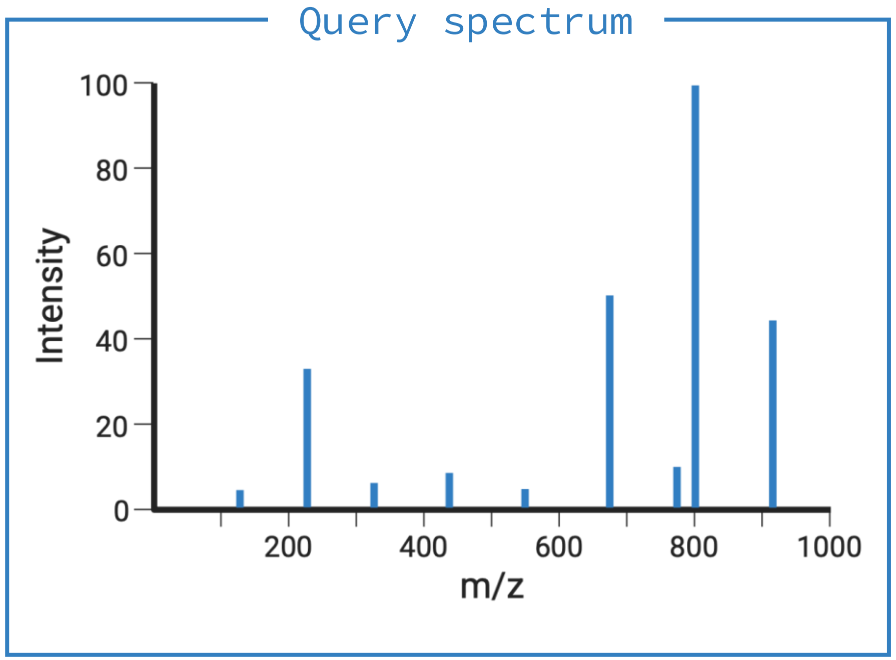
Spectral library matching

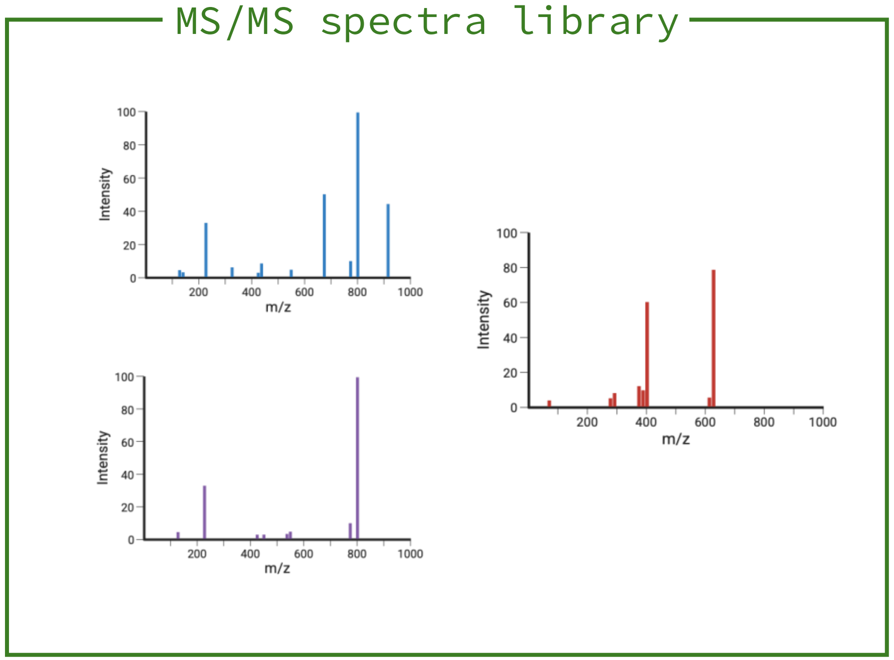


Metabolomics Interpretation

Peak annotation using LC-MS^2 spectra

Spectral library matching

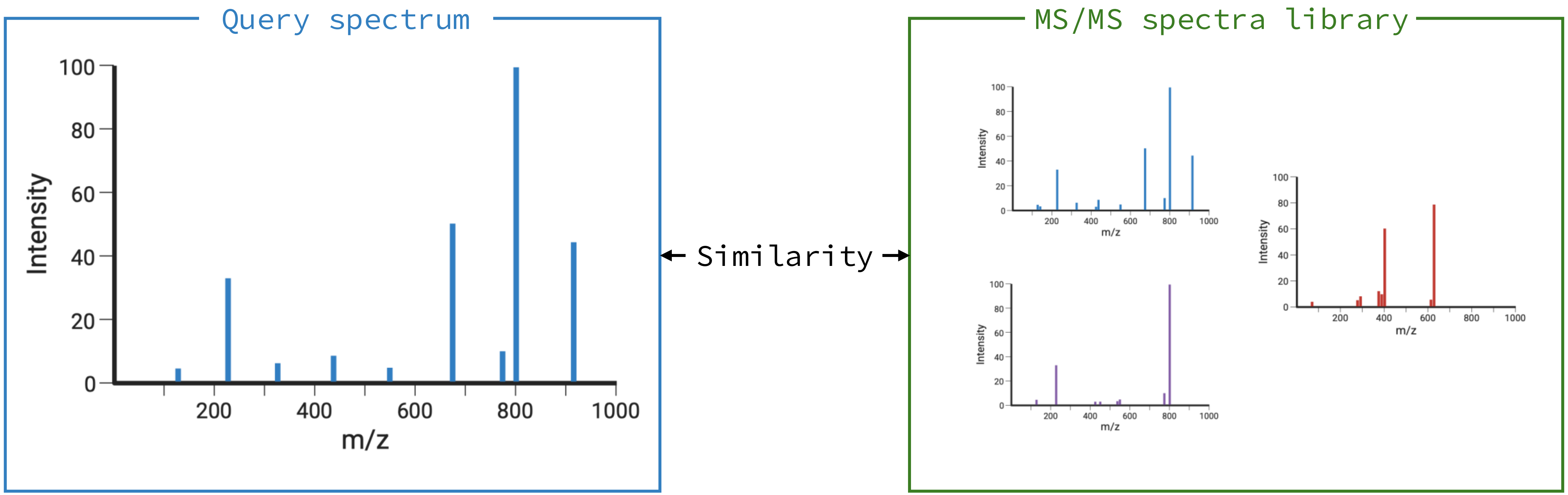




Metabolomics Interpretation

Peak annotation using LC-MS^2 spectra

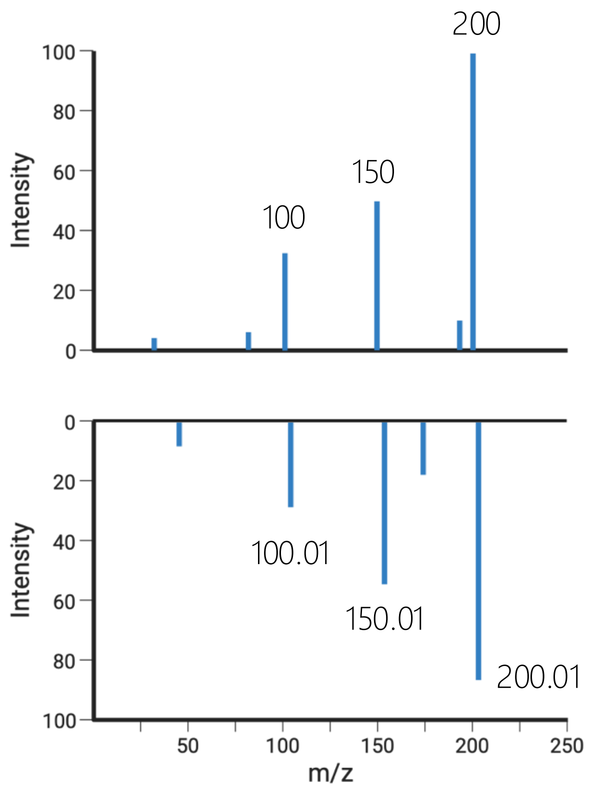
Spectral library matching



Metabolomics Interpretation

Peak annotation using LC-MS^2 spectra

Cosine similarity

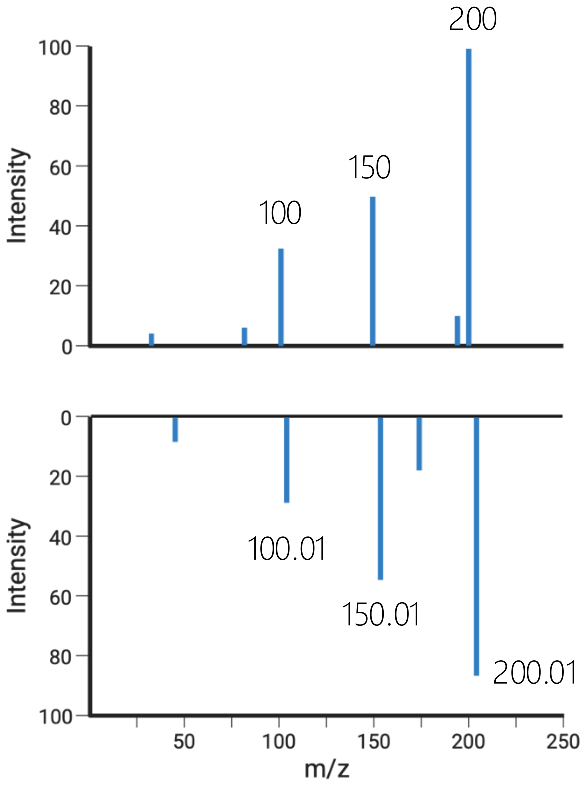


Query spectrum

Reference spectrum

Metabolomics Interpretation

Peak annotation using LC-MS^2 spectra



Cosine similarity

1. Find peaks that match: $|mz(p) - mz(p')| < t \text{ (e.g., 0.05 Da)}$

Metabolomics Interpretation

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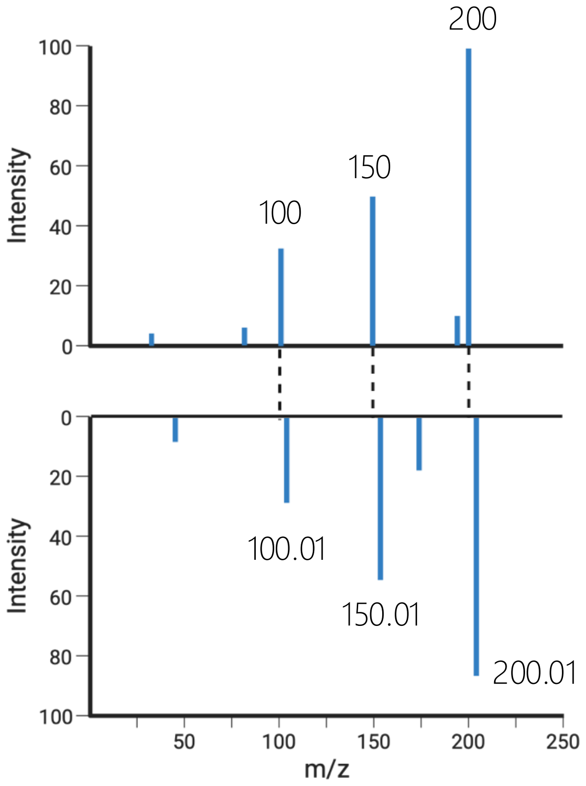

1. Find peaks that match: $|mz(p) - mz(p')| < t\ (\mathrm{e.g.}, 0.05\ \mathrm{Da})$

Metabolomics Interpretation

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Cosine similarity

1. Find peaks that match: $|mz(p) - mz(p')| < t$ (e.g., 0.05 Da)

2. Intensities of matching peaks are converted to vectors

I = [38, 46, 101]

I′ = [30, 59, 85]

Metabolomics Interpretation

Peak annotation using LC-MS^2 spectra

Cosine similarity

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3. Cosine similarity:

I' = [30, 59, 85]

$$\cos(\theta) = \frac{A \cdot B}{\|A\|\|B\|} = \frac{\sum_{i=1}^{n} A_i B_i}{\sqrt{\sum_{i=1}^{n} A_i^2}\sqrt{\sum_{i=1}^{n} B_i^2}} = 0.98$$

Metabolomics Interpretation

Peak annotation using LC-MS^2 spectra

What if … Two molecules differ by a methyl group:

Molecule A precursor = 300 Da

Molecule B precursor = 314 Da

Difference = 14 Da

Many fragments may shift +14 Da

Cosine similarity approach → No matching peaks

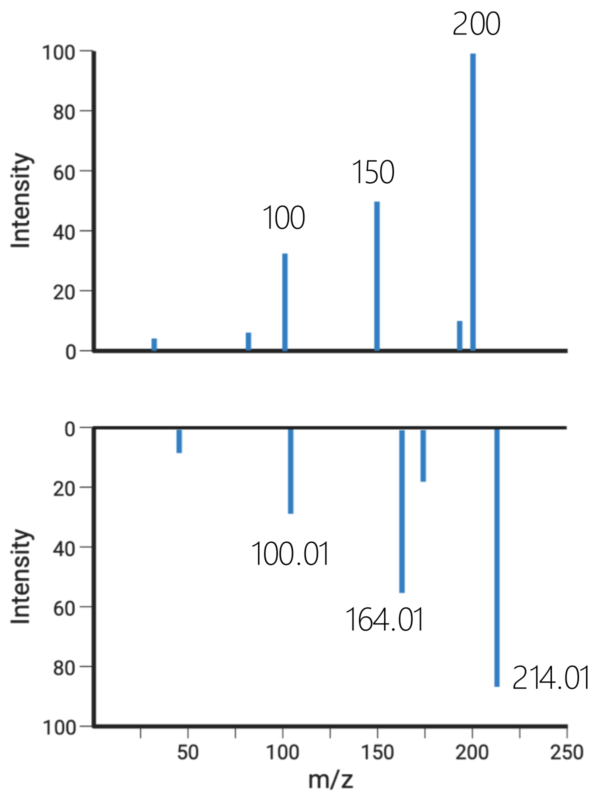
$$\text{Similarity} = 0$$

But molecules are structurally very similar… Modified cosine similarity

Metabolomics Interpretation

Peak annotation using LC-MS^2 spectra

Modified cosine similarity

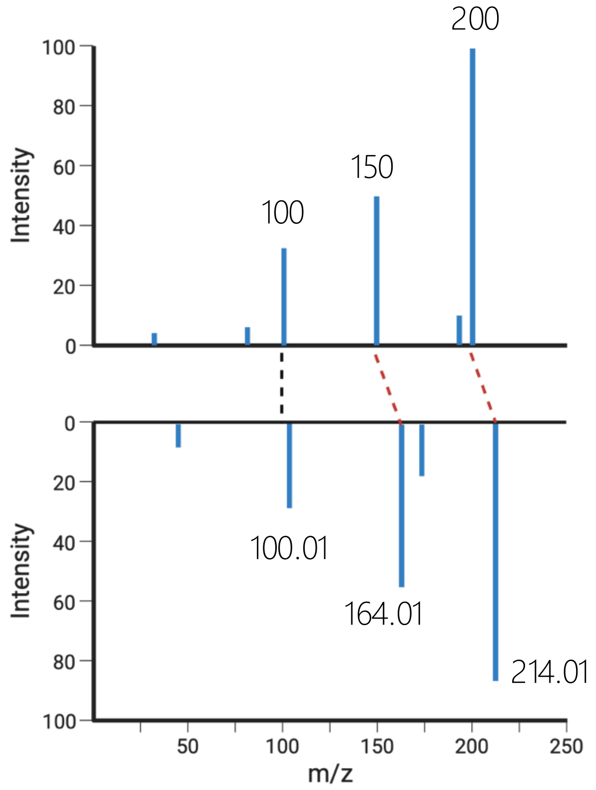


Query spectrum

Reference spectrum

Metabolomics Interpretation

Peak annotation using LC-MS^2 spectra



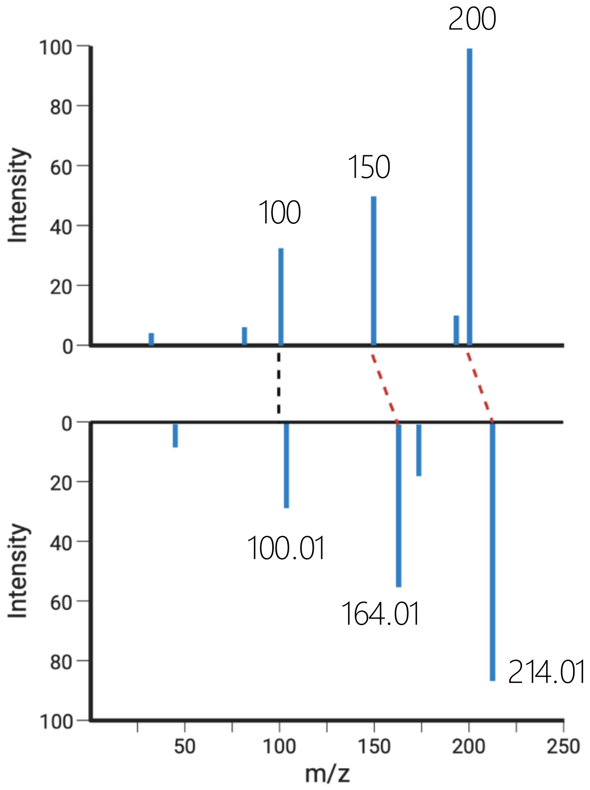
Modified cosine similarity

1. Find peaks that match: $|mz(p) + M - mz(p')| < t$

   Precursor mass difference $M = PM(S') - PM(S) = 14\text{ Da}$

Metabolomics Interpretation

Peak annotation using LC-MS^2 spectra



Modified cosine similarity

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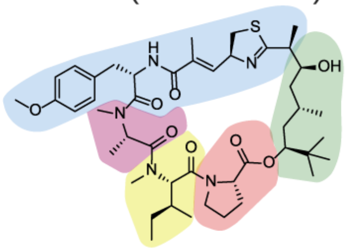
Metabolomics Interpretation

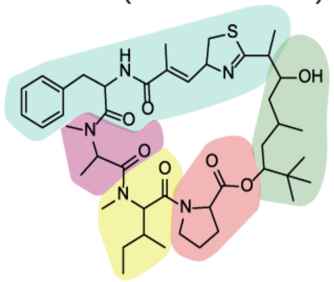
Peak annotation using LC-MS^2 spectra

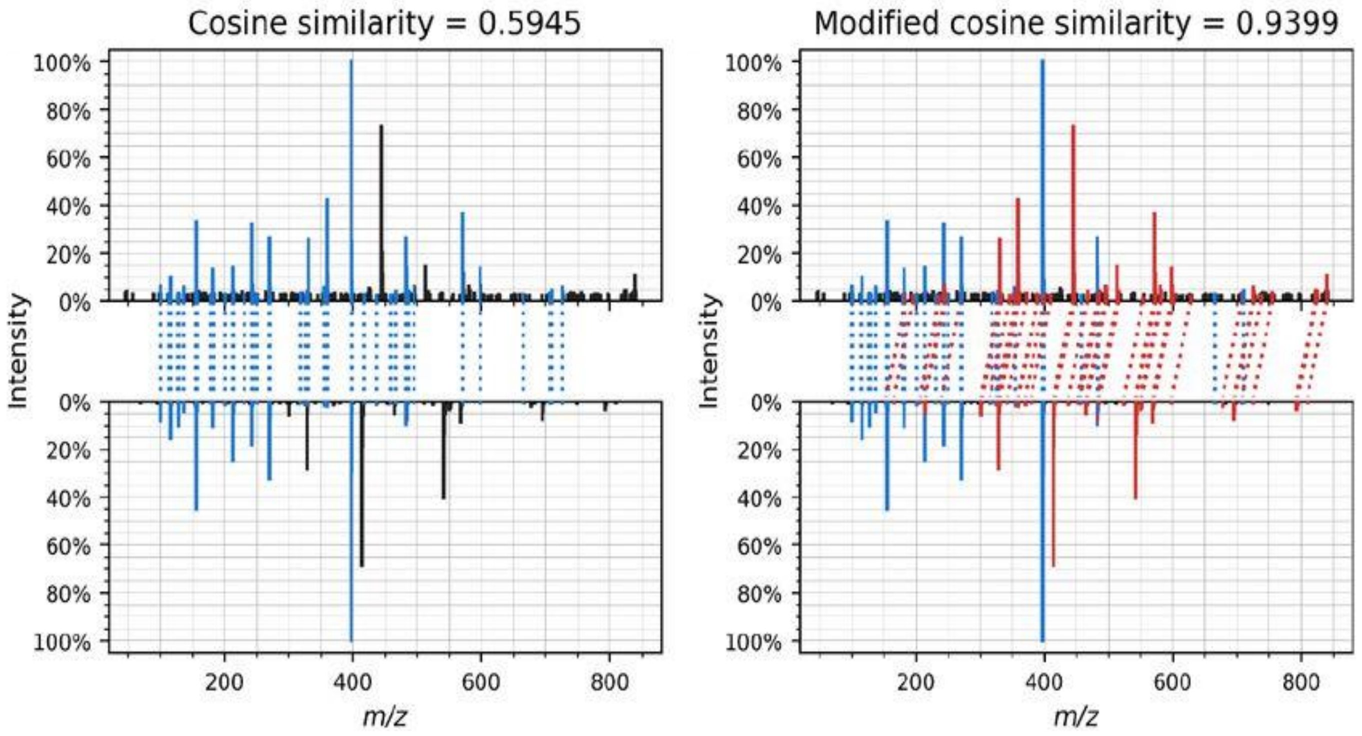
Spectral library matching

Apratoxin A (m/z = 840.497)



Apratoxin A – 30.010 Da (m/z = 810.487)





Metabolomics Interpretation

Peak annotation using LC-MS^2 spectra

Spectral library matching

However...

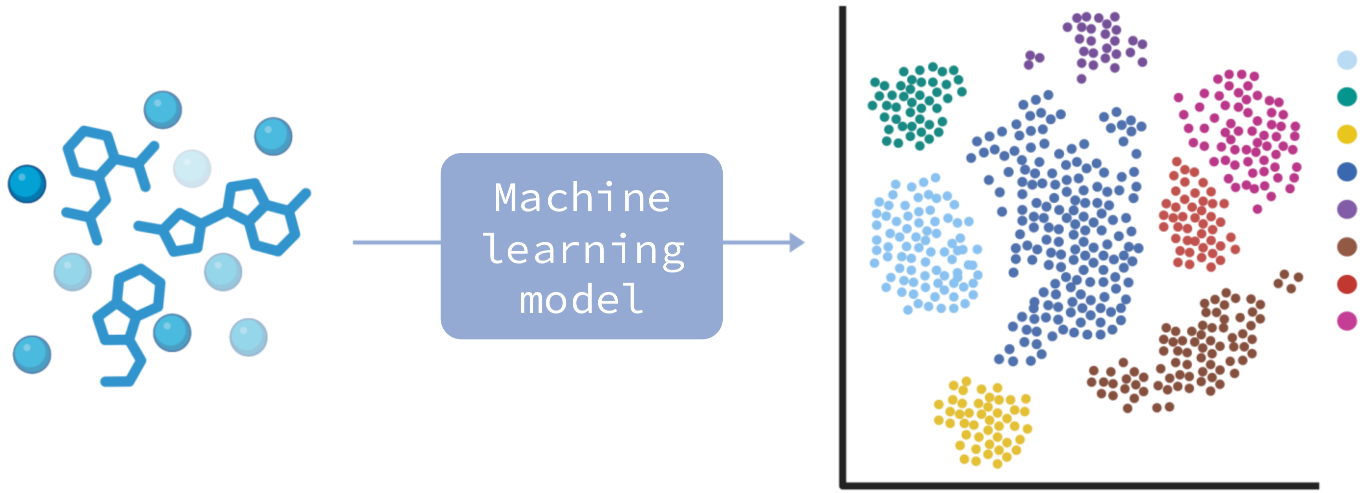
Small chemical modifications → Low modified cosine scores

Ideally, we want an approach that understands chemical structure!

Metabolomics Interpretation

Peak annotation using LC-MS^2 spectra

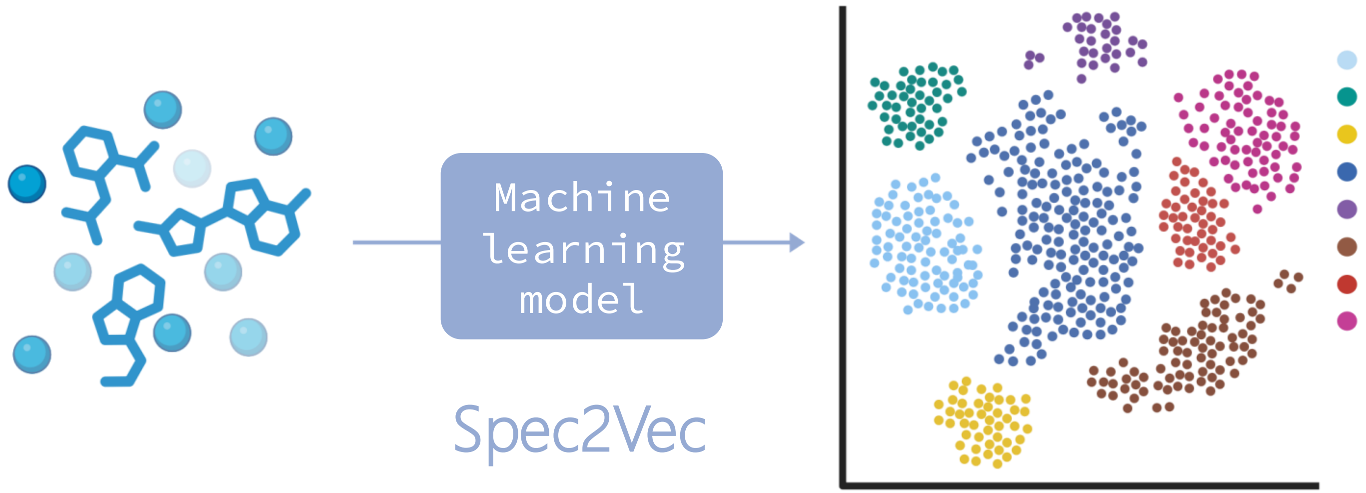
Machine learning-based similarities



Metabolomics Interpretation

Peak annotation using LC-MS^2 spectra

Machine learning-based similarities



Metabolomics Interpretation

Peak annotation using LC-MS^2 spectra

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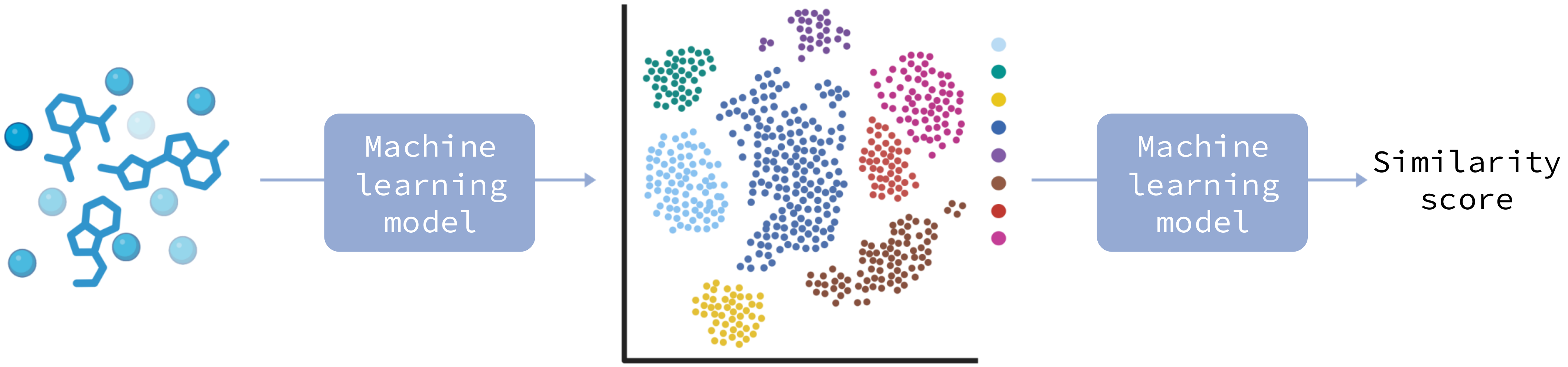

MS2Deepscore

Metabolomics Interpretation

Peak annotation with MS2Query

- **Query spectrum** – Load MS/MS spectra.
- **MS2DS** – Calculate MS2Deepscore between query spectrum and all library spectra.
- **Filter** – Select top library spectra with high MS2Deepscore.
- **Features** – Calculate features between query spectrum and selected library spectra.
- **Random Forest** – Combine features into one score using a Random Forest model.
- **Re-Rank** – Rank on Random Forest score. Select highest scoring library spectrum.

Metabolomics Interpretation

Pathway enrichment

Traditional research
One biomolecule at a time

≠

High-throughput technologies
Thousands of biomolecules

Results →

List of relevant biomolecules

Metabolomics Interpretation

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Interpretation

Extensive literature research
Lists do not reflect the complex interactions in biological systems

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Interpretation →

Extensive literature research
Lists do not reflect the complex interactions in biological systems

Functional enrichment
Focus on molecules as a group
Knowledge in public DBs

- Over-representation analysis
- Functional class scoring
- Pathway topology-based

Metabolomics Interpretation

Pathway enrichment

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≠

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Functional enrichment
Focus on molecules as a group
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Over-representation analysis

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Metabolomics Interpretation

Over-Representation Analysis (ORA)

Test if the proportion of differentially expressed molecules within a particular pathway is statistically higher or lower than would be expected by chance.

Metabolomics Interpretation

Over-Representation Analysis (ORA)

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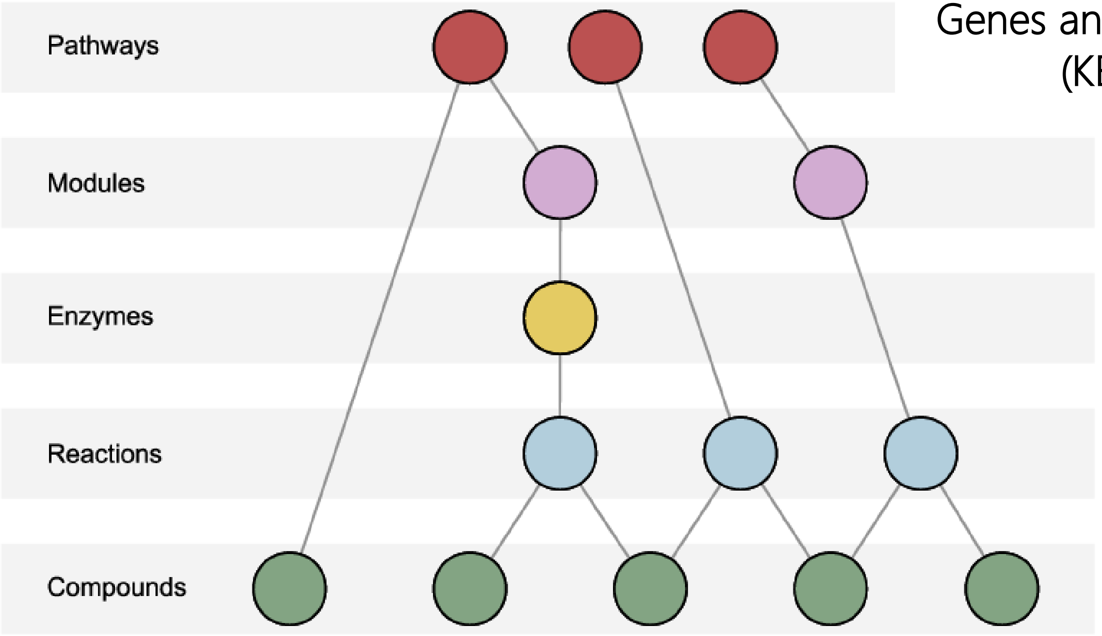

Kyoto Encyclopedia of Genes and Genomes (KEGG)

Picart-Armada, Sergio, et al. "FELLA: an R package to enrich metabolomics data." BMC bioinformatics

Metabolomics Interpretation

Over-Representation Analysis (ORA)

Background: 100 metabolites measured (that belong to 3 pathways)

Pathway	Total # of metabolites
Pathway A	10
Pathway B	20
Pathway C	30
Not in a pathway	40
Total	100

Metabolomics Interpretation

Over-Representation Analysis (ORA)

Background: 100 metabolites measured (that belong to 3 pathways)

10 differentially expressed metabolites

Pathway	Total # of metabolites	Significant metabolites
Pathway A	10	5
Pathway B	20	2
Pathway C	30	1
Not in a pathway	40	2
Total	100	10

Metabolomics Interpretation

Over-Representation Analysis (ORA)

Background: 100 metabolites measured (that belong to 3 pathways)

10 differentially expressed metabolites

How many metabolites would we expect by chance?

Pathway	Total # of metabolites	Significant metabolites	Random
Pathway A	10	5	$10 \times (10/100) = 1$
Pathway B	20	2	$10 \times (20/100) = 2$
Pathway C	30	1	$10 \times (30/100) = 3$
Not in a pathway	40	2	
Total	100	10	

Metabolomics Interpretation

Over-Representation Analysis (ORA)

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Metabolomics Interpretation

Over-Representation Analysis (ORA)

Contingency table for Pathway A

	In A	Not in A	Total
Significant	5	5	10
Not significant	5	85	90
Total	10	90	100

Hypergeometric distribution: $P(X \geq 5)$

Interpretation – Hands on

Acore enrichment functionality

Search

Acore Documentation

API usage examples

- Metabolomics data filtering
- Normalization of samples
- Batch correction of samples
- Imputation data (MS-example)
- Imputation (Metabolomics example)
- Exploratory Analysis
- Metabolomics drift correction
- Differential regulation: T-test for two groups
- Differential regulation: ANOVA across more than two groups and posthoc t-tests
- Differential regulation: ANCOVA for two groups
- **Enrichment analysis**

Enrichment analysis

Is done separately for up- and downregulated genes as it's assumed that biological processes are regulated in one direction.

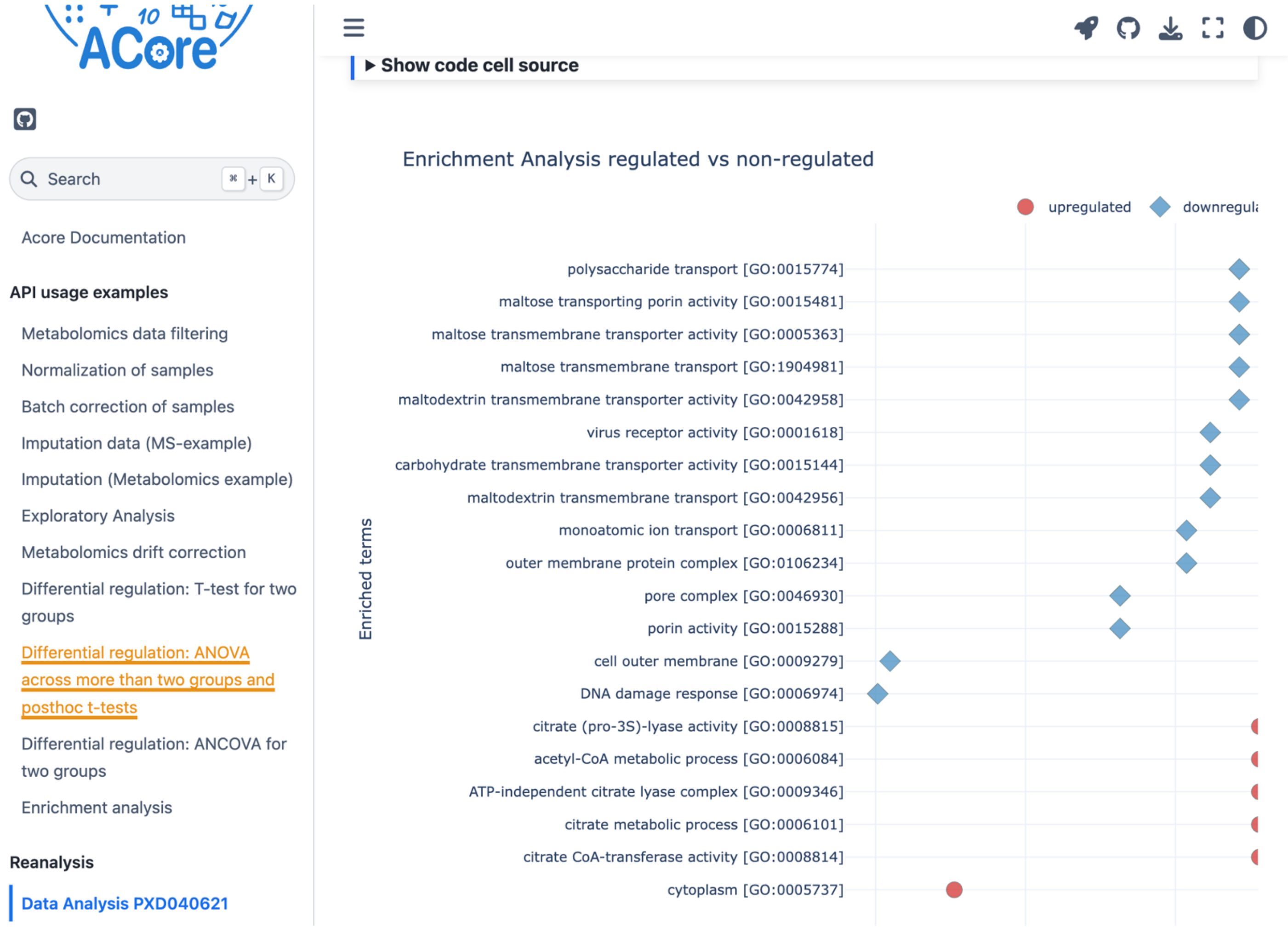
▸ **Show code cell source**

	identifier	group1	group2	pvalue	padj	rejected	log2FC	FC
98	O43432	deceased	alive	0.000	0.001	True	-0.741	0.598
32	Q96JJ3	deceased	alive	0.001	0.018	True	-0.553	0.682
99	O43175	deceased	alive	0.000	0.001	True	1.321	2.498
97	P39059	deceased	alive	0.000	0.000	True	1.600	3.031

Running the enrichment analysis for the up- and down regulated protein groups separately with the default settings of the function, i.e. a log2 fold change cutoff of 1 and at least 2 protein groups detected in the set of proteins defining the functional annotation.

```
ret = acore.enrichment_analysis.run_up_down_regulation_enrichment(
    regulation_data=diff_reg,
    annotation=annotations,
    pval_col="padj",
    min_detected_in_set=2,
    lfc_cutoff=1,
)
```

```
No significant enrichment found with the given parameters. Returning an empty DataFrame.
```


ACore
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API usage examples
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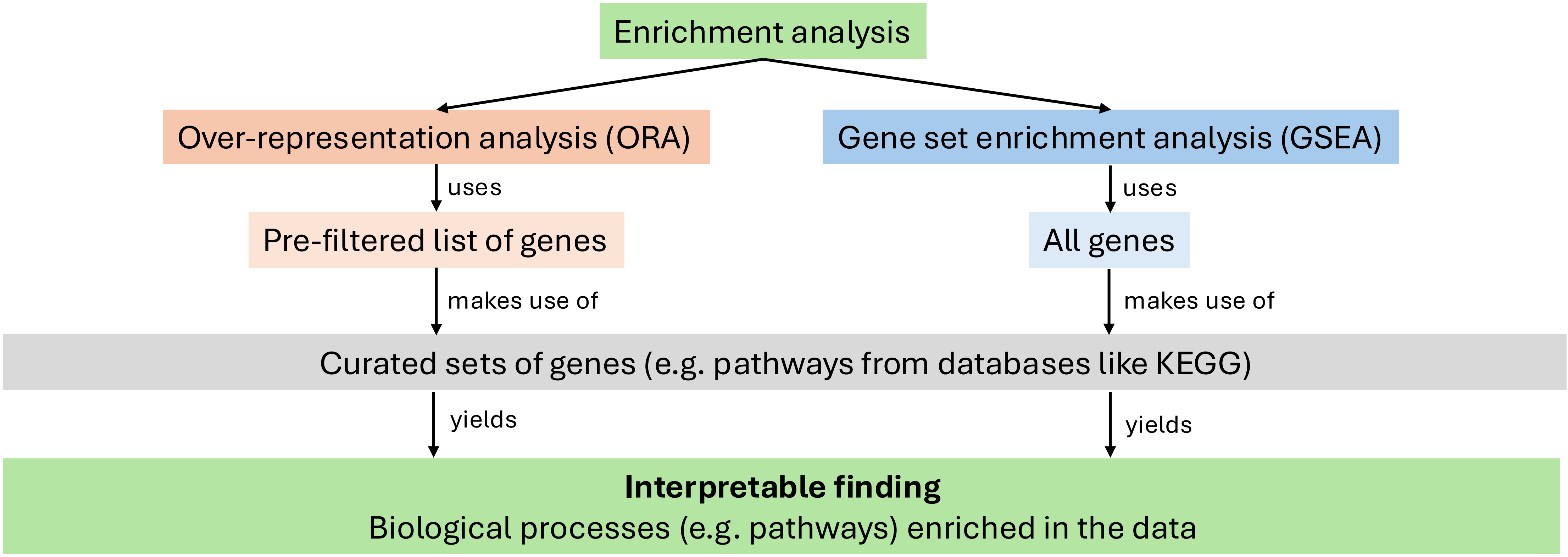
Enrichment analysis
Reanalysis
Data Analysis PXD040621
Show code cell source
Enrichment Analysis regulated vs non-regulated
upregulated
Enriched terms
polysaccharide transport [GO:0015774]
maltose transporting porin activity [GO:0015481]
maltose transmembrane transporter activity [GO:0005363]
maltose transmembrane transport [GO:1904981]
maltodextrin transmembrane transporter activity [GO:0042958]
virus receptor activity [GO:0001618]
carbohydrate transmembrane transporter activity [GO:0015144]
maltodextrin transmembrane transport [GO:0042956]
monoatomic ion transport [GO:0006811]
outer membrane protein complex [GO:0106234]
pore complex [GO:0046930]
porin activity [GO:0015288]
cell outer membrane [GO:0009279]
DNA damage response [GO:0006974]
citrate (pro-3S)-lyase activity [GO:0008815]
acetyl-CoA metabolic process [GO:0006084]
ATP-independent citrate lyase complex [GO:0009346]
citrate metabolic process [GO:0006101]
citrate CoA-transferase activity [GO:0008814]
cytoplasm [GO:0005737]

Some slides from transcriptomics course

Functional analysis for biological interpretation

- **Enrichment analysis**
  - Knowledge-based approach for biological interpretration of omics data, e.g. RNAseq results
  - **Goal:** find biological terms (curated gene sets for biological processes) that are enriched in the data for biological interpretation (no single-gene interpretation)
  - Biological term/pathway lists are obtained from KEGG, GO etc or can be manually curated (often .gmt files)
  - Different types of enrichment analysis



Functional analysis for biological interpretation

- **Gene-set enrichment analysis (GSEA)** – <u>all genes</u> ranked by fold-change and p-value

List of **<u>all</u>** genes

	Gene symbol	p-val	Differentially expressed	FC
1	***HAO1***	**0.0001**	**Yes**	**-5.6**
2	*RDH16*	0.00045	Yes	-7.6
3	*ALDOB*	0.00009	Yes	-6.7
4	*AKR1C4*	0.0051	Yes	[illegible]
5	*ABCB11*	0.067	Yes	[illegible]
6	*BAAT*	0.0034	Yes	23.5
7	*SLC2A2*	0.00990	Yes	2
8	*FCER1G*	0.00673	Yes	6.7
		...		
6532	*OSM*	0.00009	Yes	9
6533	*VEGFA*	0.067	Yes	8.5
6534	*ZAP70*	0.0034	Yes	7.8

$$\text{RANKING} = \text{SIGN(FC)} * -\log_{10}(\text{p-value})$$

Ranked list of genes

	Gene symbol	Ranking
1	*JAK9*	12.9
2	*RDH16*	11.4
3	*ALDOB*	11.3
5	*AKR1C4*	10.7
6	*ABCB11*	8.6
7	*BAAT*	3.6
8	*SLC2A2*	0.0099
9	*FCER1G*	0.00673
	...	
6532	*MyD88*	-9
6533	*TRIFF*	-9.37
6534	*IRAK1*	-15.6

SIGNIFICANT + UPREGULATED

NON-SIGNIFICANT

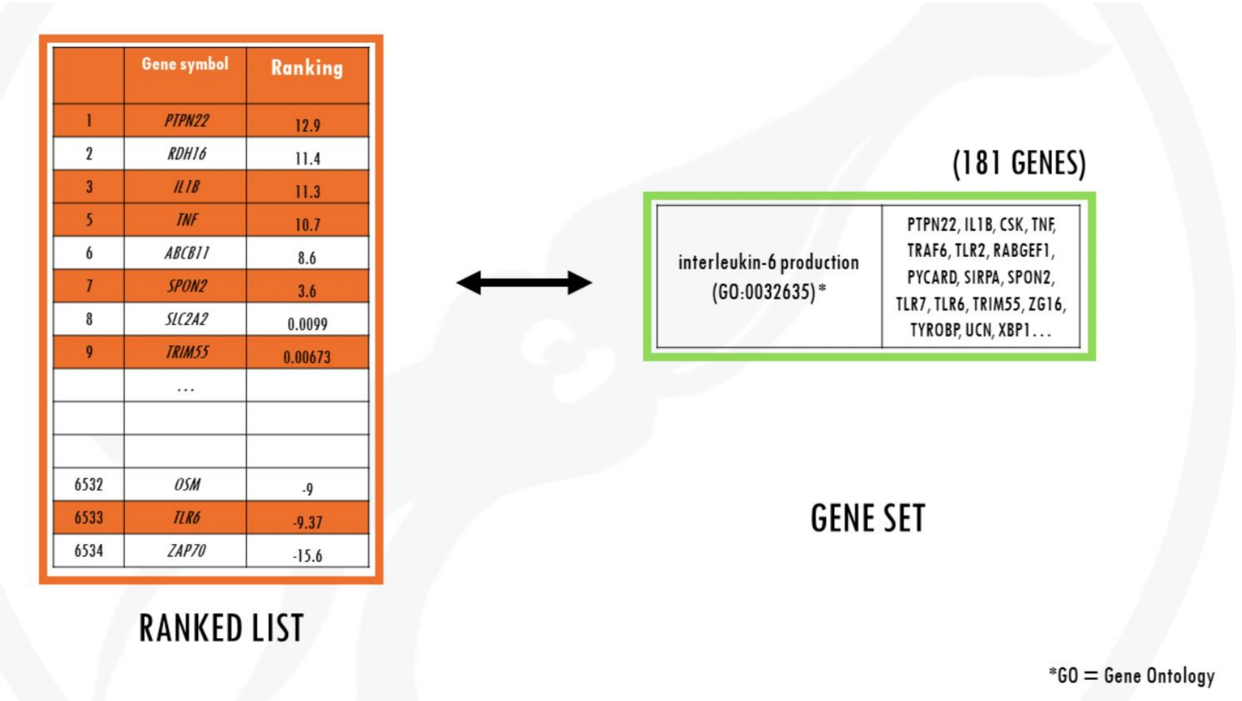
SIGNIFICANT + DOWNREGULATED

[Modified from available illustrations](#).

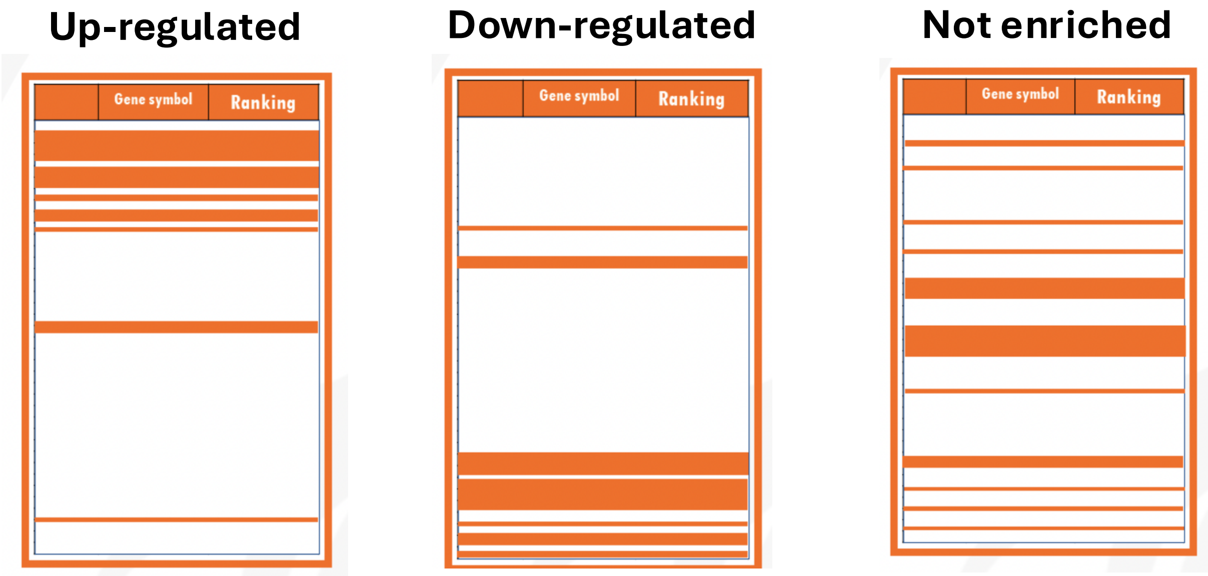
https://www.pnas.org/doi/10.1073/pnas.0506580102

Functional analysis for biological interpretation

- **Gene-set enrichment analysis (GSEA)** – Which biological processes are enriched at the top and at the bottom of the ranked gene list?



Result will show genes and biological terms that are



Statistical metrics:

- Einrichment score (ES) of a biological term represents the running sum when walking down the list
- Normalised enrichment score (NES): ES is normalised to to make biological terms with different number of genes comparable
- p-value from permutations (adjusted)

[Modified from available illustrations](https://www.pnas.org/doi/10.1073/pnas.0506580102).

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Functional analysis for biological interpretation

- **Enrichment analysis – details and summary**

	Over-representation analysis (ORA)	Gene set enrichment analysis (GSEA)
Input gene list	Filtered list, e.g. $p_{adj}<0.05$ and log2FC>|1| → Genes not passing the thresholds will be missed	Entire gene list ranked by fold-change (with sign) and/or p_{adj} → No pre-filtering of genes is an advantage
Separation of up- and down-regulated genes	Can be done	No
Core statistical result and tests used	• Typically an adjusted p value • Fisher's exact test (one-sided), hypergeometric distribution	• Normalized enrichment score (NES) and adjusted p value • Modified Kolmogorov-Smirnov and permutation test
Software packages (selection)	mulea, g:profiler, Enrichr, clusterProfiler, GSEApy (via Enrichr)	fgsea, clusterProfiler, GSEApy